

V93000 Application Note

The Wave Scale Wideband Handbook

Version 1.2.1 / SmarTest rev. 8.4.4.2

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October 10th, 2021

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0. Document History

Date	Version	Description	Author
8/1/21	1.0	Initial release	Stefan Groß
8/3/21	1.1	Corrections and enhancements based on feedback from J. Kelly and R. K. Lee	Stefan Groß
8/13/21	1.2	Minor corrections and enhancements	Stefan Groß
10/21/21	1.2.1	Minor corrections and enhancements	Stefan Groß

1. The Wave Scale RF family of instruments for the V93000 test system

Wave Scale RF systems consist of a base system and optional extensions to increase the RF carrier frequency and the modulated bandwidth or to reduce phase noise.

1.1 RF8: The Base

The RF8 base system provides RF carriers from 10 MHz to 8 GHz and modulation source and measure capabilities with up to 200 MHz modulation bandwidth. Its 4 independent subsystems provide 32 I/O ports per card. This system is suitable for 4G LTE, 5G NR FR1 (sub-6 GHz), Wi-Fi, Bluetooth and IoT testing.

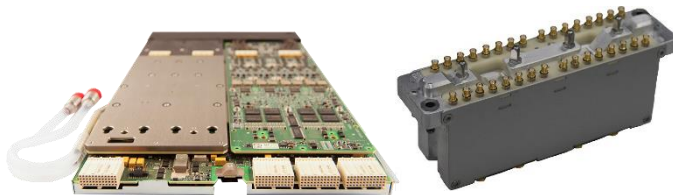


Figure 1: WSRF base system with RF8



1.2 RF18: RF extension to 18 GHz

To cover the RF frequency range to 5.85 GHz to 18 GHz the RF interface module RF8 can be replaced by the RF interface module RF18. Even though the RF18 itself has 2 GHz bandwidth, the modulation source remains with 200 MHz modulation bandwidth. RF18 provides 16 RF ports per card and up to 32 per test system.

In combination with a Wave Scale wideband (WSWB) instrument it provides the RF frequencies and the modulation bandwidth required for UWB testing.

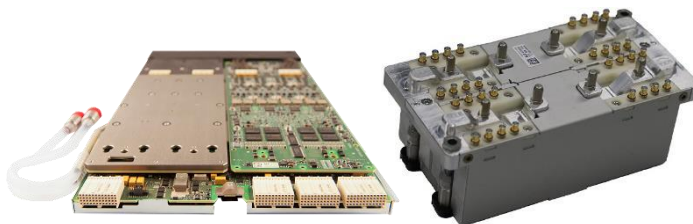


Figure 2: WSRF system with RF18

1.3 Millimeter: RF extension to 70 GHz

The water-cooled millimeter wave card cage covers RF frequencies from 24 GHz to 53 GHz or from 57 GHz to 70 GHz. The millimeter wave card cage requires an RF18 base system. In combination with a WSWB it provides up to 2 GHz modulation bandwidth covering the test needs of 5G NR FR2 IEEE 802.11ad and 802.11ay (WiGig). It provides up to 8 RF ports per block and a maximum of 64 ports per system.



Figure 3: Millimeter wave card cage

1.4 WSWB: Baseband extension to 2 GHz

The Wave Scale Wideband (WSWB) instrument E9750A is the optional 2 GHz modulation source & measure instrument for the Wave Scale RF system. It contains two 16 bit IF sources and one 12 bit IF digitizer.



Figure 4: Wave Scale Wideband card (WSWB)

1.5 The Wave Scale Precision Source (WSPS) card

The Wave Scale Precision Source (WSPS) card extends the capabilities of the WSRF card for applications that require extremely good phase noise performance for both stimulus and measurement. It is connected to the WSRF base system with the N2601P interconnect cable.

2. The WSWB instrument (E9750A)

2.1 Overview

The WSWB instrument E9750A is a channel slot wideband waveform generator that provides and captures wideband modulated signals in conjunction with an RFIM-18 and an RMM-44, RMM53 or RMM-72.

Its two 16 bit IF sources and the 12 bit digitizer with 2 GHz bandwidth support all common test items of 5G NR, WiGig and UWB devices, like modulated power, spectral mask, EVM, NMSE, IP3, LO feedthrough or residual sideband test. The AWG and the digitizer operate independently and fully parallel.

WSWB transmits and receives the complex baseband waveform on a variable intermediate carrier called base frequency. It connects to the AUX ports of the RFIM-18 with a fixed wiring scheme.

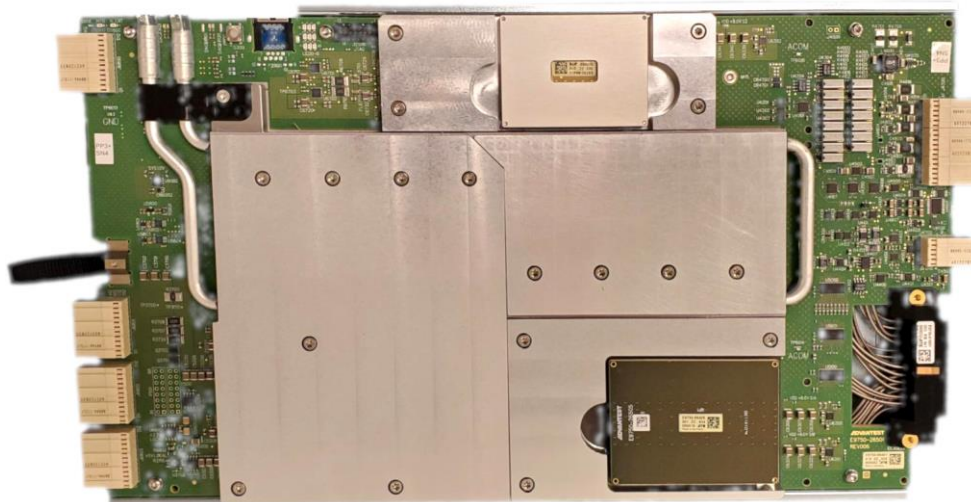


Figure 5: E9750A (WSWB) card

The instrument is controlled by an FPGA, which also contains the equalization filters. The two RF DACs feed the signal through a reconstruction filter and a ranging stage to their dedicated output ports TXA and TXB. The digitizer channel has four inputs. One low pass and two bandpass filters carry out the anti-alias filtering.

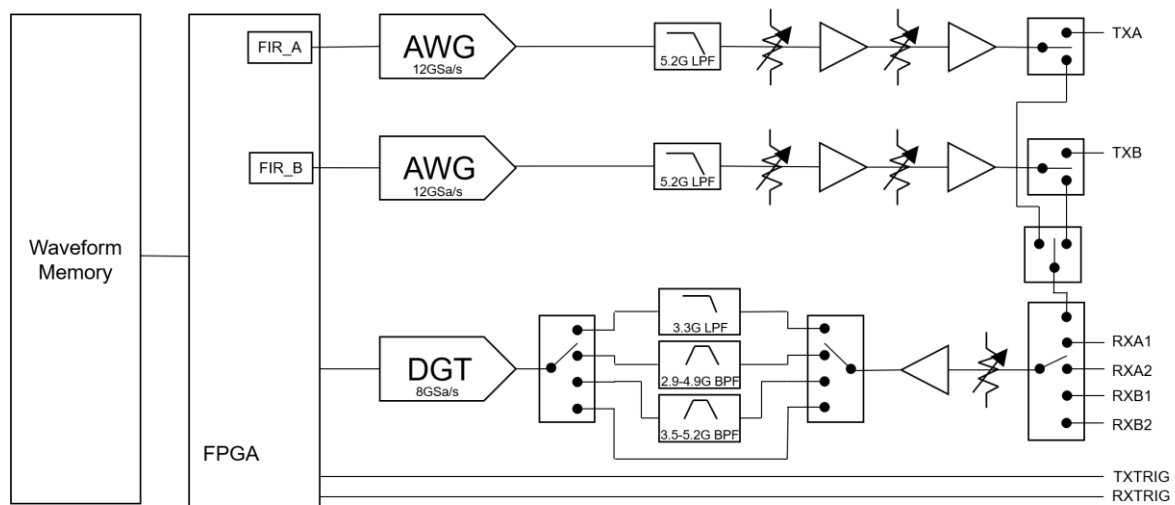


Figure 6: WSWB block diagram

2.2 Ethernet Connection

WSWB communicates with the workstation via an ethernet connection. Advantest 93000 test systems with one or more WSWB cards require a workstation equipped with the network adapter Intel X710-T4 (E9750N). SmartTest configures the ethernet connection and assigns the IP address automatically. The WSWB entries in the *tester.model* file must contain the correct serial numbers of the WSWB cards for the IP configuration.

2.3 WSWB connection to the RF subsystem

WSWB is connected to the six AUX ports of the RFIM-18 interface module. These signals have to be routed over the DUT board. The first WSWB connects to first RFIM-18 (on the left hand side), a potential second WSWB connects to the second RFIM-18 (on the right hand side).

The WSWB in- and outputs connect to the following RFIM-18 in- and outputs:

WSWB	RFIM-18
TXA	TX AUX A
TXB	TX AUX B
RXA1	RX AUX A1
RXA2	RX AUX A2
RXB1	RX AUX B1
RXB2	RX AUX B2

Table 1: WSWB to RFIM-18 connection scheme

WSWB	AWG/ DIG	WSWB	RFIM-18 AUX	RFIM-18 IF	RMM
1st 231	AWG	1B TXA 23106	TX AUX A 90103	90313-90316	890101-890108, 390101-390108, 990113-990116
		1B TXB 23102	TX AUX B 90203	90413-90416	890201-890208, 390201-390208, 990213-990216
	DIG	1B RXA1 23110	RX AUX A1 90104	90301-90302	890101-890108, 990101-990102
		1B RXA2 23112	RX AUX A2 90114	90303-90304	390101-390108, 990103-990104
		1B RXB1 23114	RX AUX B1 90204	90401-90402	890201-890208, 990201-990202
		1B RXB2 23116	RX AUX B2 90214	90403-90404	390201-390208, 990203-990204
2nd 310	AWG	2B TXA 31006	TX AUX A 90903	91113-91116	190901-190908, 290901-290908, 990913-990916
		2B TXB 31002	TX AUX B 91003	91213-91216	191001-191008, 291001-291008, 991013-991016
	DIG	2B RXA1 31010	RX AUX A1 90904	91101-91102	190901-190908, 990901-990902
		2B RXA2 31012	RX AUX A2 90914	91103-91104	290901-290908, 990903-990904
		2B RXB1 31014	RX AUX B1 91004	91201-91202	191001-191008, 991001-991002
		2B RXB2 31016	RX AUX B2 91014	91203-91204	291001-291008, 991003-991004

Table 2: WSWB and connected RFIM-18 and RMM signals and pogo numbers

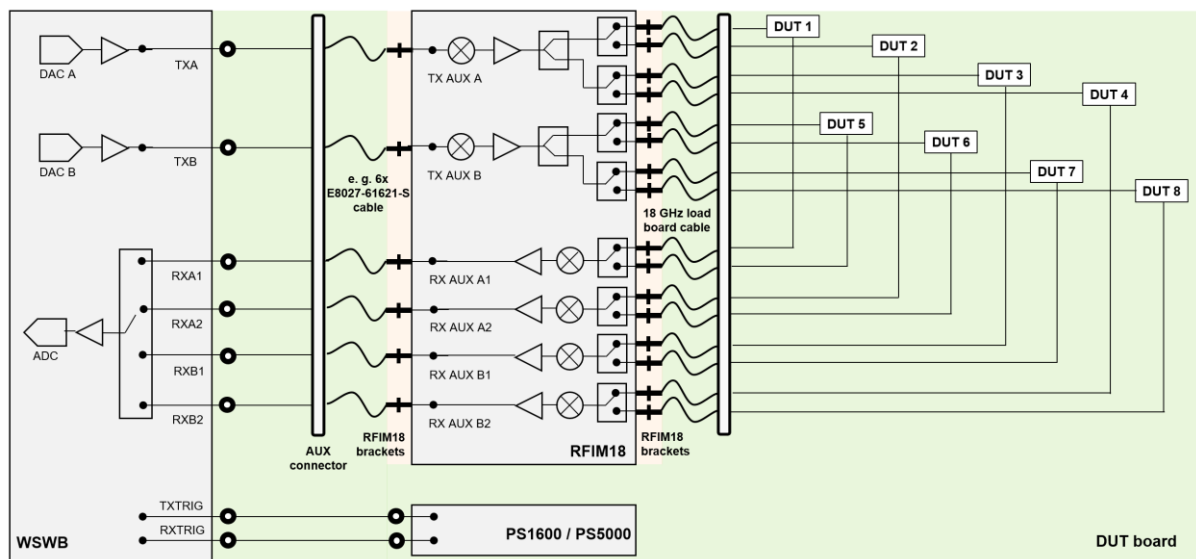


Figure 7: WSWB connected to RFIM-18

2.4 Triggering

A WSWB card requires two PS1600 or PS5000 channels for the AWG and digitizer triggering, which have to be routed over the DUT board.

3. Specification

3.1 Wave Scale Wideband AWG

Specification	Value
Resolution	16 bit
DAC sample rates	11796.48, 11968 and 11980.8 Msps
Data sample rates	<i>2G / 4G LTE / 5G NR:</i> 30.72, 61.44, 122.88, 245.76, 491.52, 983.04 and 2949.12 Msps <i>IEEE 802.11ad and IEEE 802.11ay (WiGig):</i> 1760, 2640 and 2992 Msps <i>IEEE 802.15.4 (Ultra-Wideband):</i> 499.2, 998.4 and 2995.2 Msps
AWG Memory	3 GByte (= 750M IQ samples)
Bandwidth	2 GHz complex
Filter	Low pass 5.2 GHz
Full-scale	+15 dBm +13 dBm when using flatness compensation
2nd harmonics output power <= -5 dBm (typical data)	-39 dBc @ base frequency <= 4.0 GHz -37 dBc @ base frequency > 4.0 GHz
3rd harmonics output power <= -5 dBm	-66 dBc
IMD3 output power <= 0 dBm	-50 dBc
Noise density output power <= -15 dBm	-143 dBm/Hz @ base frequency <= 3.6 GHz -145 dBm/Hz @ base frequency > 3.6 GHz
Residual EVM (loopback, typical data)	<i>5G NR ¹⁾:</i> 0.6 % @ base frequency 1.1 GHz 1.1 % @ base frequency 3.9 GHz 1.2 % @ base frequency 4.4 GHz <i>IEEE 802.11ad (WiGig) ²⁾:</i> 1.5 % @ base frequency 1.1 GHz 1.8 % @ base frequency 3.9 GHz and 4.4 GHz

Table 3: Specifications of the WSWB AWG

¹⁾ 256QAM, 400 MHz bandwidth, -18 to -2 dBm Stim Power

²⁾ 16QAM, 2 GHz bandwidth, -14 to 0 dBm Stim Power

3.2 Wave Scale Wideband Digitizer

Specification	Value
Resolution	12 bit
ADC sample rates	5222.4, 5280, 5491.2, 5898.24, 5990.4, 6000, 7864.32, 7920 and 7987.2 Msps
Data sample rates	<i>2G / 4G LTE / 5G NR:</i> 30.72, 61.44, 122.88, 245.76, 491.52 and 983.04 Msps 2611.2 and 5222.4 Msps (3.4 - 3.95 GHz base frequency only) 2949.12 and 5898.24 Msps (3.95 - 5.0 GHz base frequency only) 7864.32 Msps (500 MHz - 2.9 GHz base frequency only) <i>IEEE 802.11ad and IEEE 802.11ay (WiGig):</i> 1760 and 2640 Msps 5280 Msps (3.4 - 3.95 GHz base frequency only) 6000 Msps (3.95 - 5.0 GHz base frequency only) 7920 Msps (500 MHz - 2.9 GHz base frequency only) <i>IEEE 802.15.4 (Ultra-Wideband):</i> 499.2 and 998.4 Msps 5491.2 Msps (3.4 - 3.95 GHz base frequency only) 5990.4 Msps (3.95 - 5.0 GHz base frequency only) 7987.2 Msps (500 MHz - 2.9 GHz base frequency only)
Digitizer Memory	3 GByte (= 2G IF samples = 1G IQ samples)
Bandwidth	2 GHz complex
Filters	Low pass 3.3 GHz Band pass 2.9 to 4.9 GHz Band pass 3.5 to 5.2 GHz
Full-scale	+14 dBm (peak)
2nd harmonics input power 0 dBm, base frequency ≥ 1.8 GHz	-45 dBc
3rd harmonics input power 0 dBm	-48 dBc
IMD3 input power +6 dBm	-52 dBc @ base frequency ≤ 2.9 GHz -50 dBc @ base frequency ≥ 3.4 GHz

Residual EVM (loopback, typical data)	<i>5G NR ¹⁾:</i> 0.6 % @ base frequency 1.1 GHz 1.1 % @ base frequency 3.9 GHz 1.2 % @ base frequency 4.4 GHz <i>IEEE 802.11ad (WiGig) ²⁾:</i> 1.5 % @ base frequency 1.1 GHz 1.8 % @ base frequency 3.9 GHz and 4.4 GHz
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Table 4: Specifications of the WSWB digitizer

¹⁾ 256QAM, 400 MHz bandwidth, -18 to -2 dBm Stim Power

²⁾ 16QAM, 2 GHz bandwidth, -14 to 0 dBm Stim Power

4. Physical Interface

A 93000 test system can be equipped with a maximum of two WSWB cards. The 1st WSWB is connected to the 1st RFIM-18, which is located on the left hand side of the test head, and the 2nd WSWB is connected to the 2nd RFIM-18, which is located on the right hand side of the test head.



Figure 8: WSWB pogo cable E9750P

The WSWB instrument connects to the WSWB pogo cable E9750P, which comes with two connector blocks. Block A contains the digital trigger inputs, while block B contains the IF signals which connect WSWB to the RFIM-18 AUX ports.

The pogo block numbers are assigned as follows:

	A Digital	B IF
1 st WSWB	431	231
2 nd WSWB	311	310

Table 5: Pogo block number assignment

The signals are assigned as follows in the pogo blocks:

Pogo	Block A digital	Block B IF
1	TXTRIG	GND/TXB_POGOLOOP ¹⁾
2	SPARE	TXB
3	SPARE	GND
4	RXTRIG	GND
5	SPARE	GND/TXA_POGOLOOP ¹⁾
6	SPARE	TXA
7	SPARE	GND
8	SPARE	GND
9	NC	GND/RXA1_POGOLOOP ¹⁾
10	NC	RXA1
11	NC	GND/RXA2_POGOLOOP ¹⁾
12	NC	RXA2
13	NC	GND/RXB1_POGOLOOP ¹⁾
14	NC	RXB1
15	NC	GND/RXB2_POGOLOOP ¹⁾
16	NC	RXB2

¹⁾ only for pogo cable diagnostics

Table 6: Signal assignment of the pogo blocks

WSWB pogo connectors

The graph below shows the WSWB pogo connectors A and B with the connected signals.

Pogo connector A

	a	b	c	d
01		TXTRIG	■	o
02		■	SPARE	
03		SPARE	■	
04		■	RXTRIG	
05		SPARE	■	
06		■	SPARE	
07		SPARE	■	
08		■	SPARE	
09		NC	■	
10		■	NC	
11		NC	■	
12		■	NC	
13		NC	■	
14		■	NC	
15		NC	■	
16		■	NC	
17		NC	■	o

Pogo connector B

	a	b	c	d
01		GND	■	o
02		■	TXB	
03		GND	■	
04		■	GND	
05		GND	■	
06		■	TXA	
07		GND	■	
08		■	GND	
09		GND	■	
10		■	RXA1	
11		GND	■	
12		■	RXA2	
13		GND	■	
14		■	RXB1	
15		GND	■	
16		■	RXB2	
17		GND	■	o

legend: ■ = ground , o = orientation marker

Figure 9: WSWB pogo connectors

Millimeter Wave card cage

The millimeter wave card cage connects the WSWB RF ports to the RFIM-18 AUX ports. These ports are accessible for measurement and debug purposes via standard SMA connectors which are located above the handlebars of the millimeter wave card cage casing.



Figure 10: WSWB to RFIM-18 connectors on the Millimeter wave card cage

Four RFIM18 TX IF ports and four RX IF ports are connected to the RMMs and are not accessible. The remaining RFIM18 RF ports are accessible in the pogo blocks 9901, 9902, 9909 and 9910 in group 9.

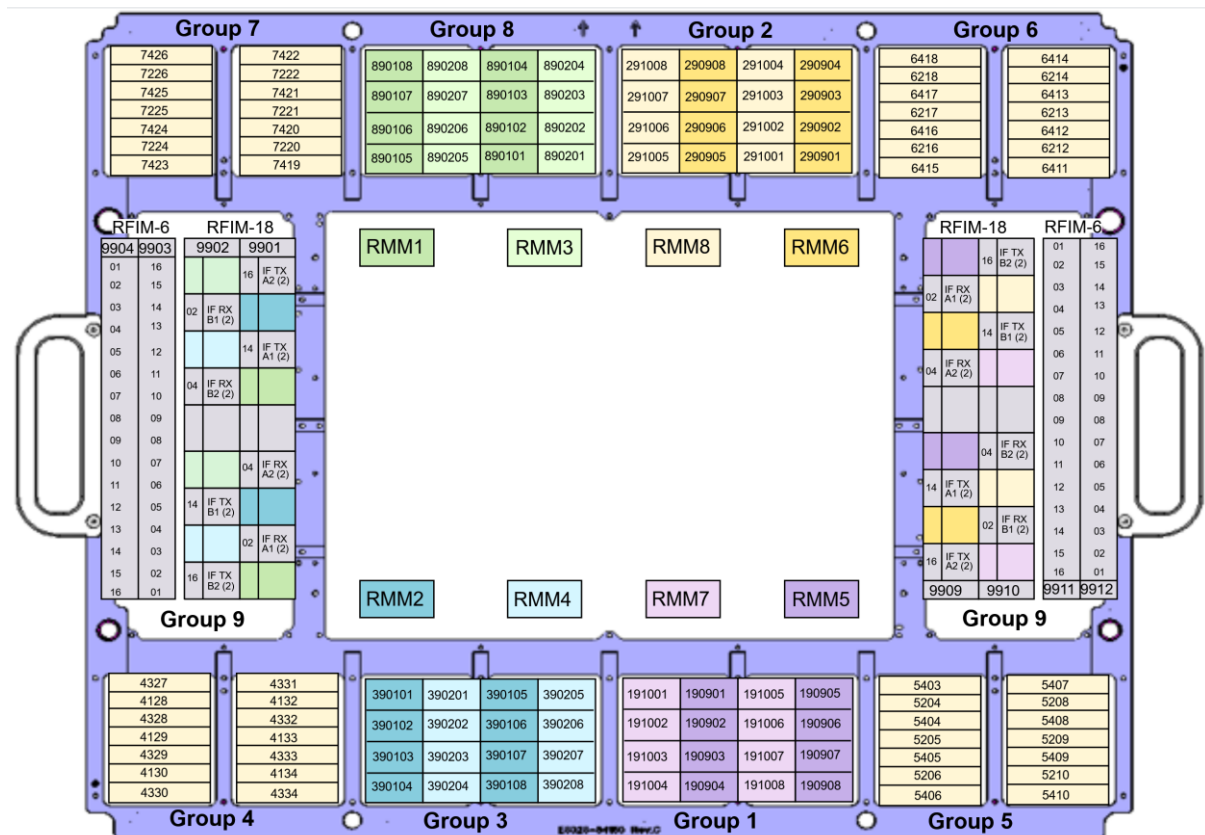


Figure 11: Wave Scale Millimeter wave card cage DUT interface

5. Setting up WSWB

5.1 Model file

The WSWB entry in the *tester.model* file contains an extension for the serial number of the WSWB card. This is required for the correct setup of the IP connection. The serial number of the card figures on the card sticker.



Figure 12: Sticker of the WSWB card with the serial number *043005017*

5.1.1 *Tester.model* file without Wave Scale millimeter card cage

The WSWB card entries in the *tester.model* file contains the hardware ID *WSWB*, two pogo block numbers and the serial number. The first pogo block number is the digital pogo block A which contains the trigger inputs. The second pogo block number is the IF pogo block B which contains the IF in- and outputs. The serial number of the card figures on the card sticker.

```
CARD_CONFIGURATION
SLOT = 104, HW = WSWB, pogoblocks = 431 231, serialNumber =
    043005017
SLOT = 106, HW = WSWB, pogoblocks = 311 310, serialNumber =
    067008761
```

Figure 13: *tester.model* file of a 93000 test system with two WSWB cards (without a millimeter wave card cage)

5.1.2 [Tester.model file with Wave Scale millimeter card cage](#)

Tester.model files for Advantest 93000 test systems with the millimeter wave card cage contain the entry for the Wave Scale millimeter DUT interface *dut_interface = WSMM*. The redirect pogo block numbers of the Wave Scale millimeter DUT interface have to be added to the entries for the PS1600 or PS5000 and the WSWB cards:

```
...
dut_interface = WSMM
...

CARD_CONFIGURATION
slot = 104, HW = WSWB, serialnumber = 043005017, pogoblocks = 431
                231, redirectpogoblocks = 7224 NA
slot = 106, HW = WSWB, serialnumber = 067008761, pogoblocks = 311
                310, redirectpogoblocks = 7228 NA
```

Figure 14: *tester.model* file of a 93000 test system with one WSWB card and a millimeter wave card cage

5.2 DUT Board

All WSWB signals are assigned to pogo numbers in the DUT board file. These entries contain extensions for the two digital trigger channels; one for the AWG and one for the digitizer. The trigger channels are shared for different sites.

The digital trigger channels **must not** be listed in the digital signals section and cannot be used or shared for other purposes. Both trigger lines must be routed over the DUT board.

```
dutboard {
  signal WSWB_DAC {
    site 1 { pogo = 23106; triggerChannel = 10101; }
    site 2 { pogo = 23102; triggerChannel = 10101; }
  }
  signal WSWB_DGT {
    site 1 { pogo = 23110; triggerChannel = 10102; }
    site 2 { pogo = 23112; triggerChannel = 10102; }
  }
  ...
}
```

Figure 15: WSWB channels in the *dutboard* file (without millimeter Wave card cage)

DUT board files of test systems with millimeter wave card cage contain the statement *redirect = true*, the identifier *testerpogo* instead of *pogo*, and six-digit pogo numbers for the trigger channels.

```
dutboard {

  redirect = true

  signal IF_RX {
    site 1 { testerpogo = 23106; triggerChannel = 520101; }
    site 2 { testerpogo = 23102; triggerChannel = 520101; }
  }
  signal IF_TX {
    site 1 { testerpogo = 23110; triggerChannel = 520102; }
    site 2 { testerpogo = 23112; triggerChannel = 520102; }
  }
  ...
}
```

Figure 16: WSWB channels in the *dutboard* file (with millimeter Wave card cage)

The WSWB outputs are 50 Ω terminated; the only valid values for the optional properties *dutImpedanceAC* and *dutImpedanceDC* are 50 Ω and 1 M Ω .

5.2.1 De-embedding

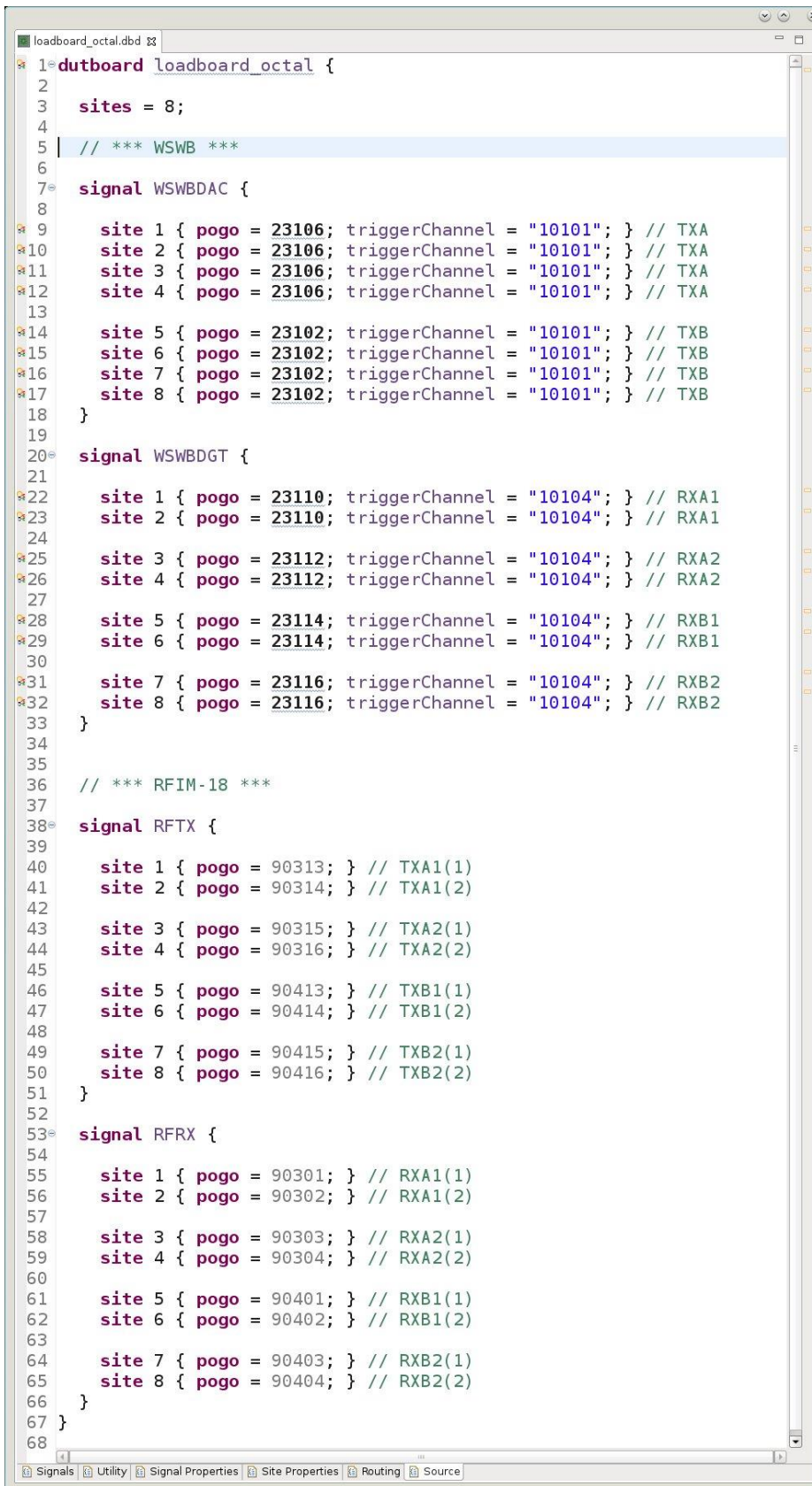
De-embedding is a technique that is used, typically to exclude the effects of the electrical paths between the DUT and the test system or, in other words, to compensate the waveform for the frequency response of the DUT board and the sockets. The de-embedding data are contained in external files such as *.citi, *.s1p, or *.s2p (touchstone) files and are usually site-dependent.

De-embedding of the DUT board or socket can be added by referencing a de-embedding file; each site of each RFIM-18 channel can reference its individual de-embedding file. In addition they can be tagged, if different files are required for different tests.

```
property deembedding {  
  rfSrc1 { // tags G0 and G1  
    G0  
    { // All sites must be assigned for each tag  
      site 1 { deembedding = deembedding/deembed_SRC1_site1_G0.citi" ; }  
      site 2 { deembedding = deembedding/deembed_SRC1_site2_G0.citi" ; }  
    }  
  
    G1  
    {  
      site 1 { deembedding = deembedding/deembed_SRC1_site1_G1.citi" ; }  
      site 2 { deembedding = deembedding/deembed_SRC1_site2_G1.citi" ; }  
    }  
  }  
}
```

Figure 17: *dutboard* file with de-embedding

The de-embedding option can also be used to equalize the stim waveform based on a focus calibration of the stim path.



```

1= dutboard loadboard_octal {
2
3   sites = 8;
4
5   // *** WSWB ***
6
7   signal WSWBDAC {
8
9       site 1 { pogo = 23106; triggerChannel = "10101"; } // TXA
10      site 2 { pogo = 23106; triggerChannel = "10101"; } // TXA
11      site 3 { pogo = 23106; triggerChannel = "10101"; } // TXA
12      site 4 { pogo = 23106; triggerChannel = "10101"; } // TXA
13
14      site 5 { pogo = 23102; triggerChannel = "10101"; } // TXB
15      site 6 { pogo = 23102; triggerChannel = "10101"; } // TXB
16      site 7 { pogo = 23102; triggerChannel = "10101"; } // TXB
17      site 8 { pogo = 23102; triggerChannel = "10101"; } // TXB
18  }
19
20  signal WSWBDGT {
21
22      site 1 { pogo = 23110; triggerChannel = "10104"; } // RXA1
23      site 2 { pogo = 23110; triggerChannel = "10104"; } // RXA1
24
25      site 3 { pogo = 23112; triggerChannel = "10104"; } // RXA2
26      site 4 { pogo = 23112; triggerChannel = "10104"; } // RXA2
27
28      site 5 { pogo = 23114; triggerChannel = "10104"; } // RXB1
29      site 6 { pogo = 23114; triggerChannel = "10104"; } // RXB1
30
31      site 7 { pogo = 23116; triggerChannel = "10104"; } // RXB2
32      site 8 { pogo = 23116; triggerChannel = "10104"; } // RXB2
33  }
34
35
36  // *** RFIM-18 ***
37
38  signal RFTX {
39
40      site 1 { pogo = 90313; } // TXA1(1)
41      site 2 { pogo = 90314; } // TXA1(2)
42
43      site 3 { pogo = 90315; } // TXA2(1)
44      site 4 { pogo = 90316; } // TXA2(2)
45
46      site 5 { pogo = 90413; } // TXB1(1)
47      site 6 { pogo = 90414; } // TXB1(2)
48
49      site 7 { pogo = 90415; } // TXB2(1)
50      site 8 { pogo = 90416; } // TXB2(2)
51  }
52
53  signal RFRX {
54
55      site 1 { pogo = 90301; } // RXA1(1)
56      site 2 { pogo = 90302; } // RXA1(2)
57
58      site 3 { pogo = 90303; } // RXA2(1)
59      site 4 { pogo = 90304; } // RXA2(2)
60
61      site 5 { pogo = 90401; } // RXB1(1)
62      site 6 { pogo = 90402; } // RXB1(2)
63
64      site 7 { pogo = 90403; } // RXB2(1)
65      site 8 { pogo = 90404; } // RXB2(2)
66  }
67 }
68

```

Figure 18: octal site *dutboard* file

6. AWG Setup

The WSWB AWG supports the SmarTest8 AWG instrument model. To source a wideband modulated RF signal, a *sourceWaveform* action for the WSWB and an *rfStim* action for the RFIM-18 or RMM have to be set up.

6.1 The sampling concept of the AWG

Waveforms are provided as complex baseband waveforms in I/Q format. These are up-sampled by the SmarTest software from their original sample rate to sample rates close to 3 Gsps, and then equalized by a complex FIR filter. A further by-4 up-sampler resamples the waveform to sample rates close to 12 Gsps, before a digital NCO upconverts it with the base frequency. This waveform is then converted by a D/A-converter and filtered by a 5.2 GHz analog low pass reconstruction filter.

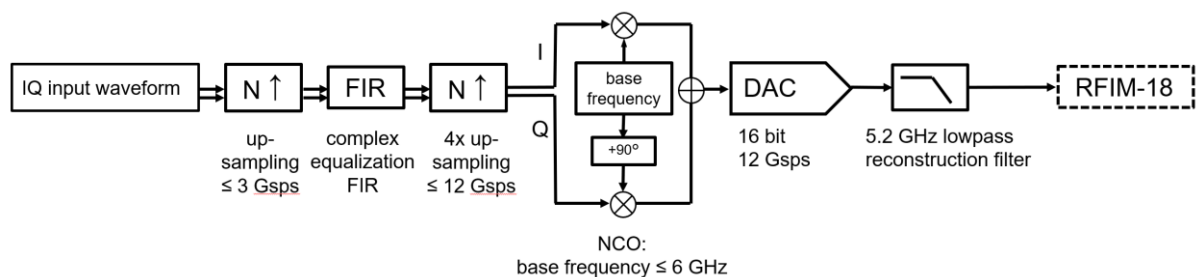


Figure 19: The WSWB AWG sampling concept

6.2 AWG instrument properties

Instrument properties apply to all actions of a pin or pin group within one specification and cannot be changed during a operating sequence. They are located inside the *setup* block of the specification file.

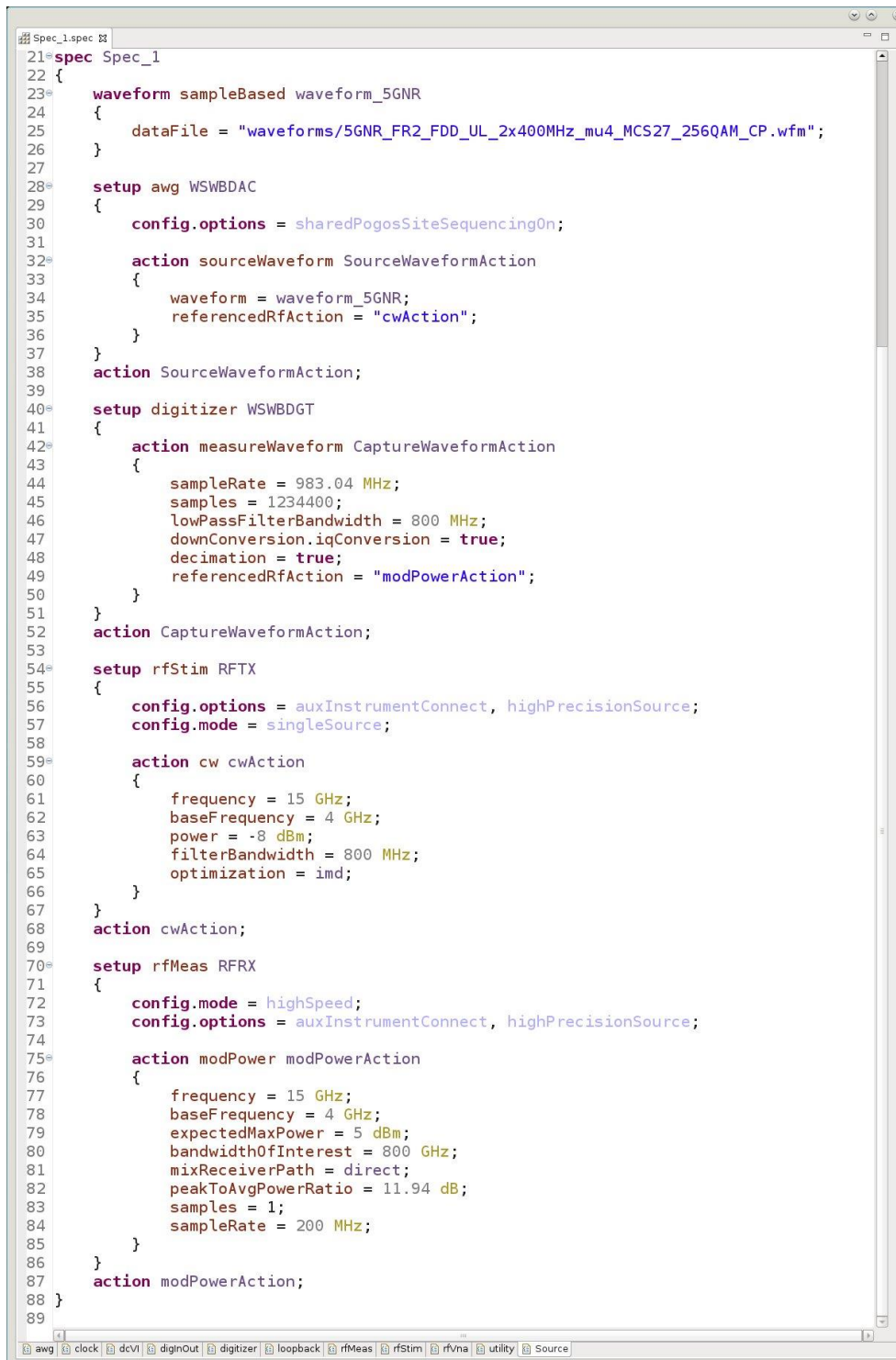


Figure 20: SmarTest8 specification file

6.2.1 `connect`

By default, an analog instrument is connected right before its first action and only disconnected at the end of the test program. If *connect* is set to *false*, the instrument will be disconnected before the execution and will stay disconnected during the execution. If *connect* is set to *true*, the instrument is connected before the start of the action, and remains connected after the measurement.

Default: *true*

Example:

```
connect = false;
```

6.2.2 `disconnect`

By default, an analog instrument remains connected once it has been connected, and is only disconnected at the end of the test program. Use this property to disconnect the AWG instrument after the measurement execution by setting *disconnect* to *true*. If set to *false*, the instrument remains connected after the measurement.

Default: *false*

Example:

```
disconnect = true;
```

6.2.3 `delay`

To fine-tune the synchronization between the AWG and the other instruments the actual start of an action can be delayed with the *delay* property. Use this property to adjust the phase of the signal or to add a small delay up to 1 ms. For larger delays up to 10 s and to delay single actions you can use the *wait* statement in the operating sequence.

Default: 0 ns

Example:

```
delay = 9.1ns;
```

6.2.4 `amplitudeRange`

This AWG property is ignored.

6.2.5 `superPeriod`

This property sets a super period to align all signals in this instrument's phase domain.

Default: LCM of all periods.

Example:

```
superPeriod = 103.1ns;
```

6.2.6 keepAlive

WSWB does not support the keep-alive option of the AWG instrument.

6.2.7 compensation

The standard system calibration performs frequency response measurements of the WSWB, RFIM-18, RMM44, RMM53 and RMM72 signal paths. By selecting the compensation quality *flatness* and by making use of the setup property *referencedRfAction* and the entire signal path from the WSWB AWG to the RF pogo is being equalized and flattened.

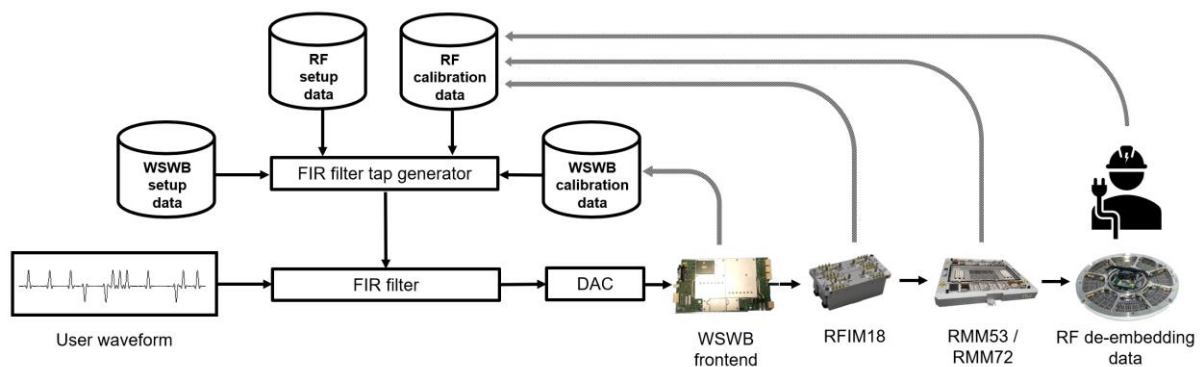


Figure 21: Equalization concept of the RF subsystem

Use the property *compensation* to set the quality of the power compensation. Supported values are *standard* (default) and *flatness*. *Standard* compensation corrects the power at the base frequency without a wideband flatness correction. *Flatness* corrects the power at the base frequency and in addition activates the wideband flatness compensation with the equalization filter.

Example:

```
compensation = flatness;
```

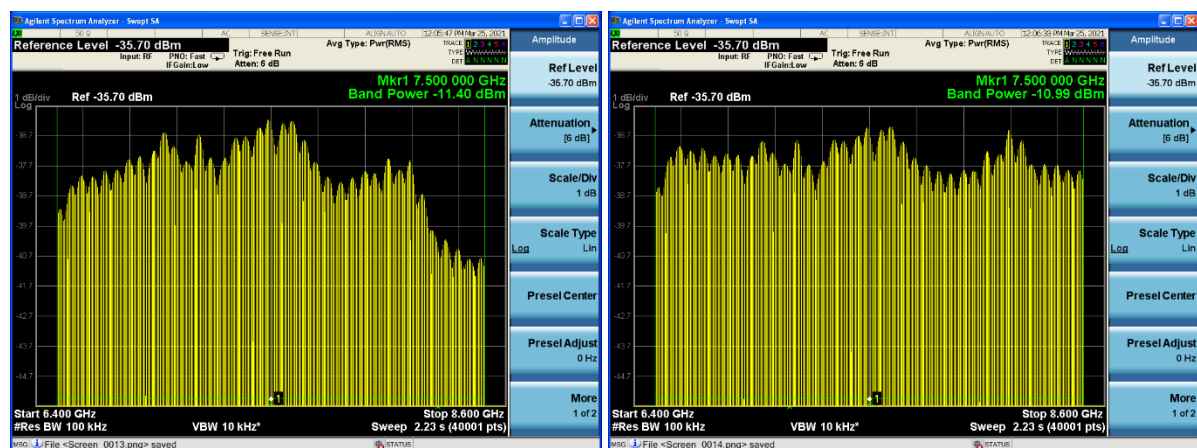


Figure 22: Wideband flatness without (left) and with flatness compensation (right picture)

6.3 AWG instrument options

Instrument properties and instrument options apply to all actions of a pin or pin group within one specification and cannot be changed during an operating sequence. In the specification file instrument options are located inside the *setup* block together with the instrument properties.

6.3.1 Differential connection

All WSWB channels are single-ended. If the AWG instrument option *differential* is selected, an error message is issued.

6.3.2 IQ option

The WSWB AWG is a pure IF source and ignores the IQ instrument option. WSWB source waveform data are complex baseband waveforms in I/Q format. WSWB up-samples and upconverts this waveform to the base frequency with a digital I/Q modulator and sources the IF signal.

6.3.3 Internal Cross-Connect

Use the property *internalCrossConnect* to internally connect the AWG output to the digitizer input without a DUT board connection and also bypassing the RFIM-18 connection. Internal cross-connect is always single-ended. To establish the connection, the option must be set for both the AWG and the digitizer.

Example:

```
config.options = internalCrossConnect;
```

6.3.4 Ground sense

This AWG instrument option is not supported. If the AWG instrument option *ground sense* is selected, an error message is issued.

6.3.5 Multi-Site testing

The AWG outputs TXA and TXB connect to the RFIM-18 inputs TX AUX A and TX AUX B. Each of them drives a buffer with two outputs, followed by multiplexer switches, so that these two inputs drive eight RFIM-18 IF outputs in total, four at a time. TXA and TXB are not independent and cannot drive different waveform data.

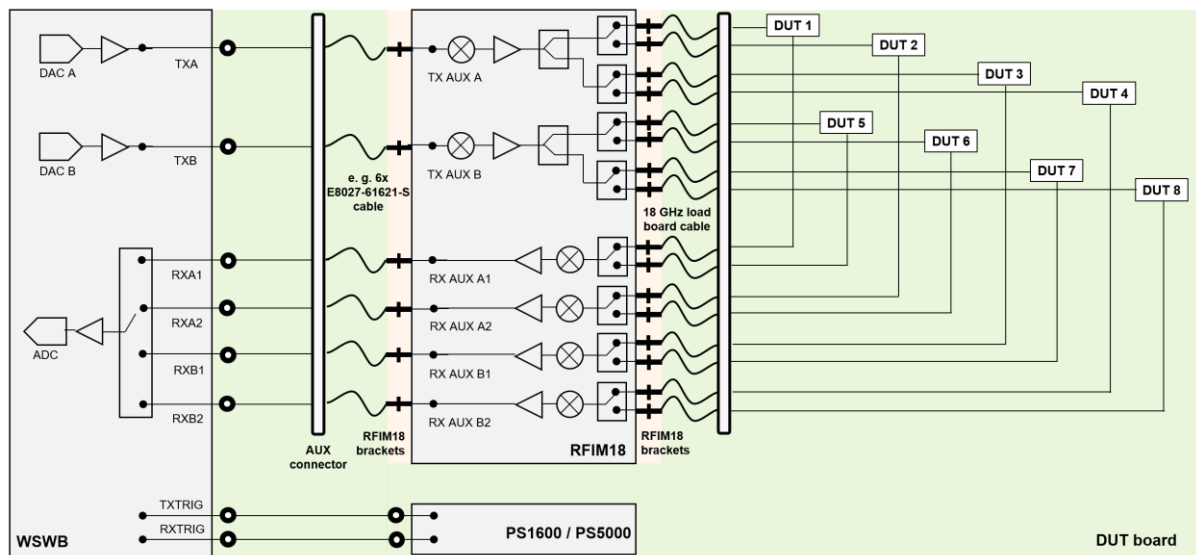


Figure 23: WSWB connected to RFIM-18

Octal site testing is supported without further switches or fan-out buffers on the DUT board with four sites at a time truly parallel.

6.3.5.1 *siteInterlacing*

The instrument option *siteInterlacing* is not supported.

6.3.5.2 *sharedPogosSiteSequencing*

The instrument option *sharedPogosSiteSequencing* executes all sites sequentially, even though they could run in parallel. It is recommended to apply this option only when compensation quality *flatness* is selected. In this case it is even mandatory to select the option, as flatness has to be compensated for each site individually.

Example:

```
config.options = sharedPogosSiteSequencingOn;
```

6.4 Analog Waveforms

Waveform data for the *awg*, *rfStim* and *dcVI* instruments are contained in the waveform specification. They contain the complete signal description including the timing information (sample rate). WSWB only supports sample-based, complex (I/Q) waveforms in the formats plain ASCII (.cwf), signal studio (.wfm) and 89600 (.vsa).

Plain ASCII waveform files consist of an optional header section with metadata, and a data section with the actual sample values. The waveform length must be a multiple of 8 and at least 256 IQ sample pairs. The format of the numbers within a waveform file is arbitrary.

The sample rate is either part of the waveform specification or the waveform file header. If both values are set, the value from the specification is taken. The WSWB AWG supports the following AWG sample rates:

2G / 4G LTE / 5G NR:

30.72, 61.44, 122.88, 245.76, 491.52, 983.04 & 2949.12 Msps

IEEE 802.11ad and IEEE 802.11ay (WiGig):

1760, 2640 and 2992 Msps

IEEE 802.15.4 (Ultra-Wideband):

499.2, 998.4 & 2995.2 Msps

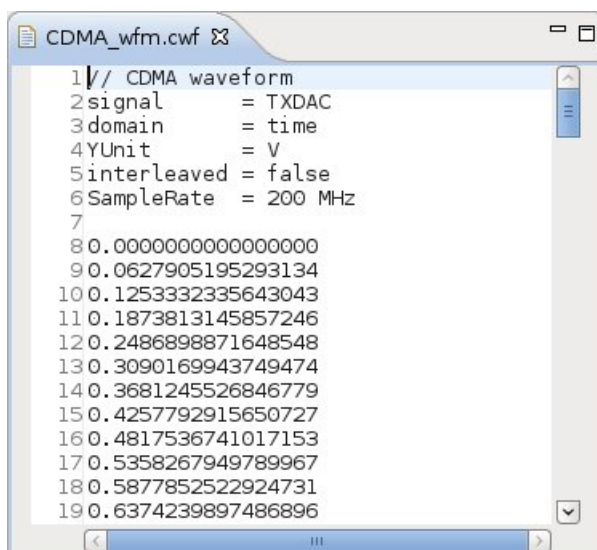


Figure 24: Custom waveform file

Please be aware that the DC content of all waveforms sourced by AWGs is removed, and the AC content is rescaled to the amplitude set by the action property *amplitude*.

Sample size and rate, and the repeat count define the duration of sample-based waveforms. Set the property *repeatInfinite* to *true* to keep the AWG running until the start of a new parallel block within the operating sequence or the end of the operating sequence.

6.5 Properties of the *sourceWaveform* action

Action properties are located inside the *action* block of the specification file. They are only effective for one specific action.

6.5.1 Waveform

Use the property *waveform* to reference the waveform if the complex waveform is contained in a single file. Use the properties *waveformI* and *waveformQ* if the real and the imaginary part are contained in separate files.

Examples:

```
waveform = lte34;           waveformI = lte34_I;
                             waveformQ = lte34_Q;
```

6.5.2 Sample Rate

The property *sampleRate* is either part of the waveform specification or the waveform file header. If both values are set, the value from the specification is taken. The WSWB AWG supports the following AWG sample rates:

2G / 4G LTE / 5G NR:

30.72, 61.44, 122.88, 245.76, 491.52, 983.04 & 2949.12 Msps

IEEE 802.11ad and IEEE 802.11ay (WiGig):

1760, 2640 and 2992 Msps

IEEE 802.15.4 (Ultra-Wideband):

499.2, 998.4 & 2995.2 Msps

6.5.3 Repeat Count

Use the property *repeatCount* to increase the number of waveform executions. The waveform of a periodic signal can be reduced to a single or a few signal periods, which are then repeated. Use this feature to save waveform memory in the WSWB instrument. Make sure that the waveform contains an exact integer number of cycles.

Default: 1

This property is mutually exclusive with the property *repeatInfinite*.

Example:

```
repeatCount = 10;
```

6.5.4 Repeat Infinite

The property *repeatInfinite* is useful to adapt the duration of a stimulus automatically to that of a measurement. With this property set to *true*, the waveform is repeated until a new parallel block is started within the operating sequence, or till the end of the operating sequence. The waveform is repeated at least until all other actions or patterns in the operating sequence have finished. This option can slightly increase the duration of the operating sequence.

Default: *false*

This property is mutually exclusive with the property *repeatCount*.

Example:

```
repeatInfinite = true;
```

6.5.5 Upsampling

The SmarTest software and the D/A converter of the WSWB AWG up-sample the user waveform to a sample rate close to 12 Gsps in two steps: The workstation up-samples the waveform to a sample rate close to 3 Gsps, and the D/A converter contains a fix x4 up-sampler.

The AWG property *upsampling* is optional; the values *auto*, *memoryOptimized* and *imageReduction* are supported. An error is returned if set to *off*; the upsampler is an integral part of the sampling concept of the WSWB AWG and can't be turned off.

Example:

```
upsampling = auto;
```

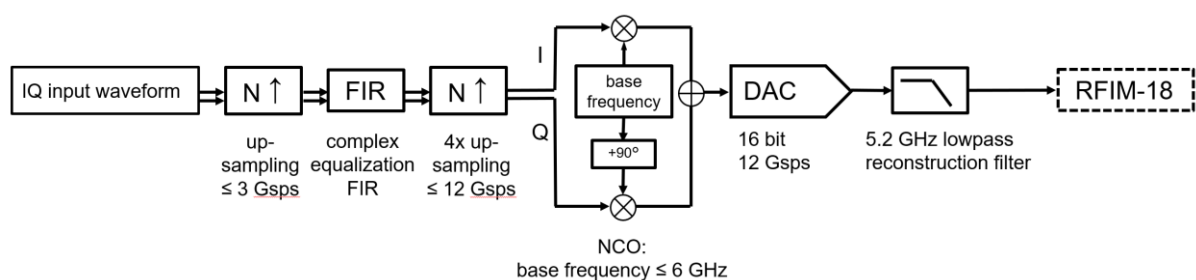


Figure 25: WSWB upsampling

6.5.6 Base frequency

The WSWB AWG output signal is a carrier signal at the *base frequency*, modulated by the complex baseband waveform. The modulation shifts the baseband waveform, which is centered around 0 Hz, to the *base frequency*, which is in the range of 545 MHz to 5.2 GHz. The property *baseFrequency* sets this carrier frequency.

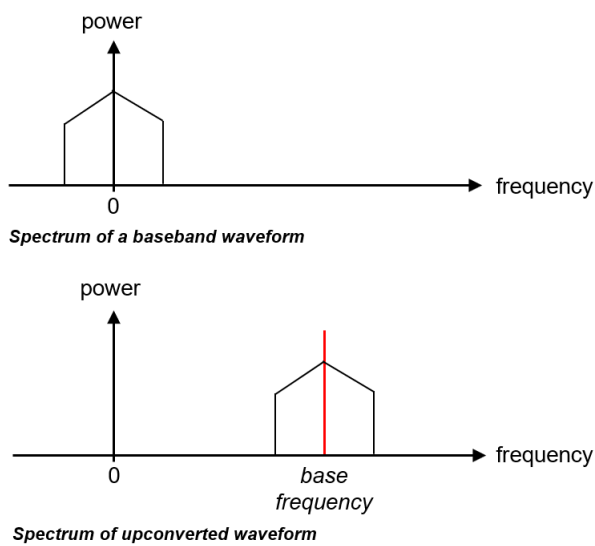


Figure 26: Base frequency

The property *baseFrequency* must not be set when *referencedRfAction* is used; in this case it is synchronized with the *cw* action setup of the *rfStim* instrument.

6.5.7 Reconstruction filter

The output of the WSWB AWG is band-limited with an analog low-pass filter, called reconstruction filter, to remove images, harmonics and high frequency noise. It has a cut-off frequency of 5.2 GHz.

The optional property *bandwidth* checks if the upconverted wideband signal is located within the passband of the filter.

Default: 1 GHz

Example:

```
bandwidth = 800 MHz;
```

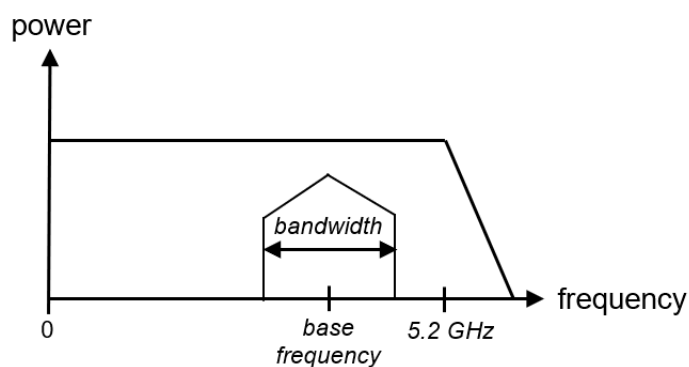


Figure 27: Upconverted signal within the 5.2 GHz reconstruction filter

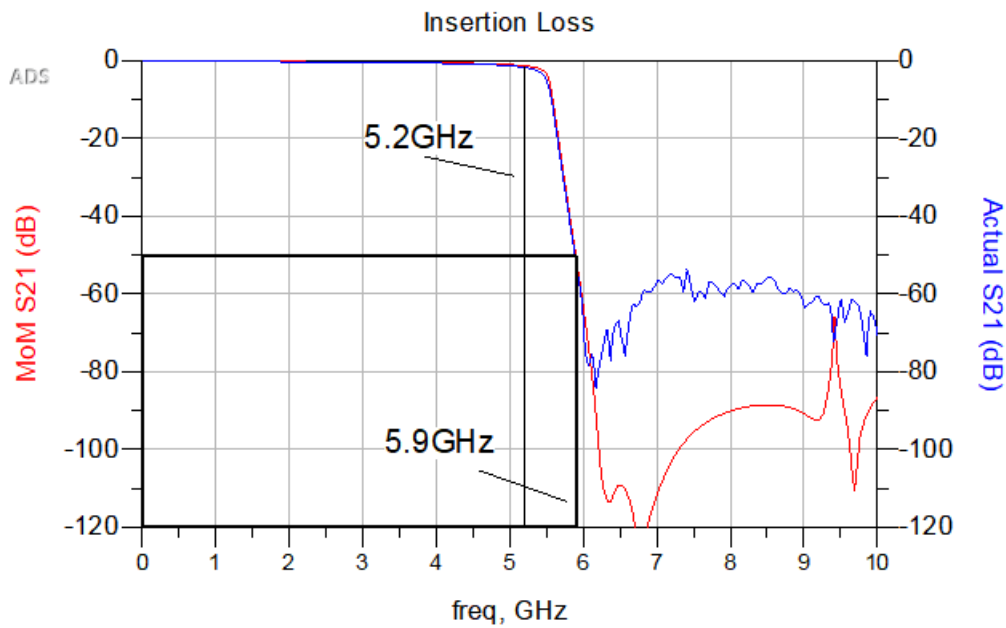


Figure 28: 5.2 GHz reconstruction filter: insertion loss (10dB/div)

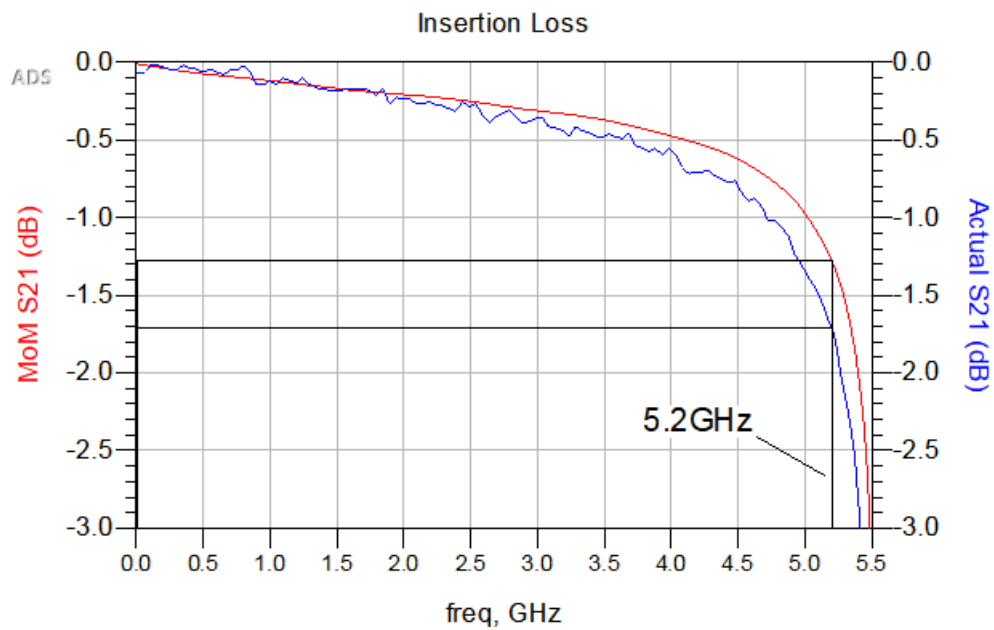


Figure 29: 5.2 GHz reconstruction filter: insertion loss (0.5dB/div)

The property *bandwidth* must not be set when *referencedRfAction* is used; in this case it is synchronized with the *cw* action setup of the *rfStim* instrument.

6.5.8 Amplitude and Power

Use the property *power* to set the average output signal power in dBm, or alternatively use the property *amplitude* to set the average output amplitude of the signal in Volts peak (V_p). These properties are mutually exclusive.

Default: power = 0 dBm

Example:

Power = -10.1 dBm;

The power of an AC waveform can either be set by the property *power* or the property *amplitude*. All mixed-signal or RF signal powers, driven by Wave Scale AWGs or captured by Wave Scale digitizers, are average powers, and the unit is dBm. All amplitudes of signals, driven by Wave Scale AWGs or captured by Wave Scale digitizers, are defined as peak amplitudes, and the unit is Volts peak (V_{peak}). This corresponds to the length of the complex phasor and is commonly used in telecommunications.

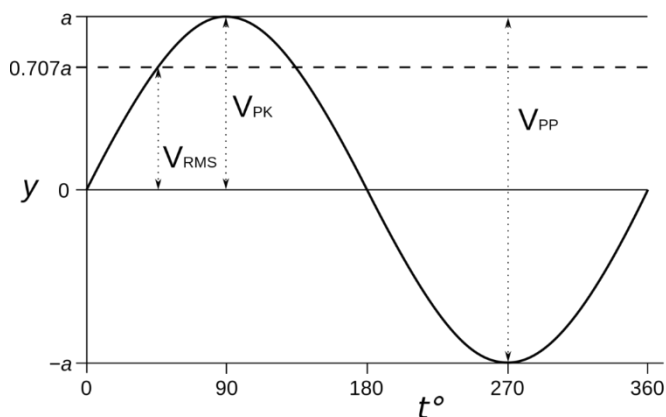


Figure 30: The amplitude of an AWG signal

The appropriate range is selected by the system software, such that the D/A converter operates with the maximum resolution. The property *amplitudeRange* is ignored.

The properties *amplitude* and *power* must not be set when *referencedRfAction* is used; in this case it is taken from the *cw* action setup of the *rfStim* instrument.

The level accuracy of the entire setup is heavily influenced by the connection between the AWG and the DUT. The power level at the DUT input may differ significantly from the power level setup value, caused by impedance mismatch and the losses and frequency responses of cables and DUT board traces. Only a thorough calibration of the entire signal path in combination with the high level repeatability of the AWG ensures a high level accuracy.

The frequency response of the connection (traces, cables, active and passive components of the DUT board and socket) can be added by referencing a *de-embedding file* (see also chapter 5.2.1 *De-embedding*).

6.5.9 Offset

The WSWB AWG does not support offset voltages; an error is returned if the property *offset* is assigned a value other than 0.0 V.

6.5.10 Target Frequency

This property is not relevant for WSWB modules and has no effect.

6.5.11 Optimization

This option is ignored on the WSWB AWG.

6.5.12 ReferencedRfAction

The WSWB AWG is the wideband IF signal source of the RFIM-18. The power, base frequency and bandwidth settings must be synchronized between these two instruments. The easiest way to accomplish this is making use of the property *referencedRfAction*. The argument of the property *referencedRfAction* is the name of the corresponding *rfStim* action.

Flatness compensation of the RFIM-18 signal path is only accomplished when *referencedRfAction* is used. The *sourceWaveform* properties *amplitude*, *power*, *base frequency* and *bandwidth* must not be set when *referencedRfAction* is used, they are mutually exclusive.

```
// rfStim setup
    ISetupRfStim rxalSetup = ds.addRfStim("RXA1");
    ISetupRfStim.ICw TX_IF = rxalSetup.cw("rxalAction");
...

// AWG setup
...
    Source_Waveform.setReferencedRfAction("rxalAction");
...
```

Figure 31: WSWB setup with referenced RF action

6.6 AWG Setup with the Device Setup API

In addition to the SmarTest Setup Format (SSF), SmarTest's own domain specific language, AWGs and digitizers can also be set up using the Device Setup API, the Java test method interface to the instruments. The Device Setup API provides a complete set of functions to set up measurements.

The example below shows the complete WSWB AWG setup for a 5G NR EVM test.



```

1 package testmethods;
2
3 import xoc.dsa.DeviceSetupFactory;
4
11 public class EVM_5GNR_2x400MHz_TX extends TestMethod {
12     IMeasurement measurement;
13
14     @Override
15     public void setup(){
16
17         IDeviceSetup ds = DeviceSetupFactory.createInstance();
18
19         //////////////////////////////////////
20         // WSWB AWG instrument setup
21         //////////////////////////////////////
22
23         // AWG waveform
24         ISetupWaveformSampleBased waveform_5GNR = ds.addWaveformSampleBased("waveform_5GNR");
25         waveform_5GNR.setDataFile("waveforms/5GNR_FR2_FDD_UL_2x400MHz_mu4_MCS27_256QAM_CP.wfm");
26         ISetupAwb awgSetup = ds.addAwb("DAC_A");
27
28         // AWG action
29         ISetupWaveform SourceWaveformAction = awgSetup.sourceWaveform("SourceWaveformAction");
30         SourceWaveformAction.setWaveform(waveform_5GNR);
31         SourceWaveformAction.setReferencedRfAction("cwAction");
32
33         //////////////////////////////////////
34         // RFIM18 TX instrument setup
35         //////////////////////////////////////
36
37         // rfStim setup
38         ISetupRfStim rftxSetup = ds.addRfStim("IFTXA1");
39         rftxSetup.setConfigOptionAuxInstrumentConnect(); // AUX input from WSWB
40         rftxSetup.setConfigModeSingleSource();
41         rftxSetup.setConfigOptionHighPrecisionSource(); // optional WSPS card
42
43         // cw action
44         ISetupRfStim.ICw TX_IF = rftxSetup.cw("cwAction");
45         TX_IF.setFrequency(11.0E9);
46         TX_IF.setBaseFrequency(4.0E9);
47         TX_IF.setPower(-2.0);
48         TX_IF.setFilterBandwidth(800E6);
49
50         // RFIM18 optimization / gain setting
51         TX_IF.setOptimizationBalanced();
52         TX_IF.setOptimizationNoiseBandwidth(100.0E6);
53
54         //////////////////////////////////////
55         // Operating sequence
56         //////////////////////////////////////
57
58         ds.parallelBegin("MainParallelGroup");
59         {
60             ds.sequentialBegin("SequentialGroupRFIMTX");
61             {
62                 ds.actionCall(TX_IF);
63             }
64             ds.sequentialEnd();
65
66             ds.sequentialBegin("SequentialGroupAWG");
67             {
68                 ds.waitCall("0.1 us");
69                 ds.actionCall(SourceWaveformAction);
70             }
71             ds.sequentialEnd();
72         } // end_parallel
73         ds.parallelEnd();
74
75         measurement.setSetups(ds);
76
77     } // end setup()
78
79     @Override
80     public void execute() {
81         measurement.execute();
82     }
83
84 }
85
86
87
88
89
90
91
92
93

```

Figure 32: WSWB AWG setup for a 5G NR EVM test

7. Digitizer Setup

The WSWB digitizer supports the SmarTest8 digitizer instrument model. To capture a wideband modulated RF signal, a WSWB *measureWaveform* action and an RF *modPower* action have to be set up.

7.1 The sampling concept of the digitizer

The real-valued IF waveform received from the RFIM-18 at base frequency, is filtered by one of the three analog anti-aliasing filters of the WSWB digitizer's filter bank. Smartest selects the appropriate filter based on the base frequency setting. The 12 bit IF sampler samples the signal with up to 8 Gbps. A complex NCO with optional I/Q conversion down-converts the digitized signal to complex baseband, symmetrically around DC.

The captured waveform is decimated to the data sample rate set by the property *sampleRate*. It can optionally be equalized and de-embedded by a digital filter.

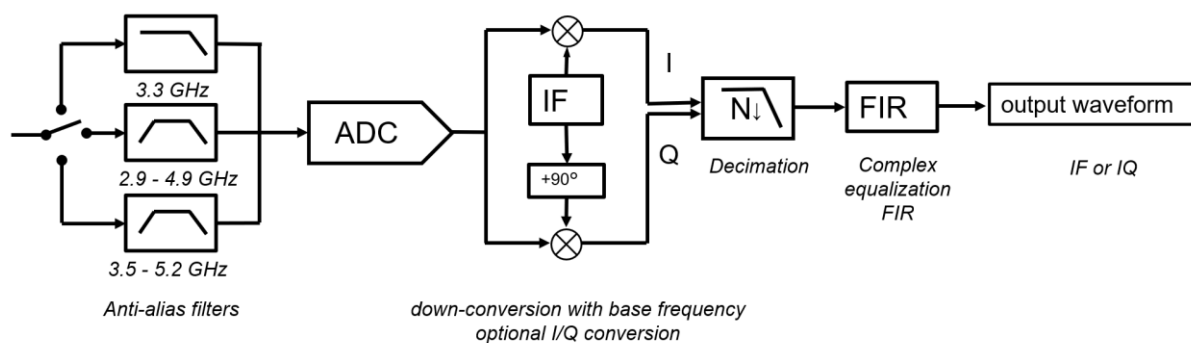


Figure 33: The sampling concept of the digitizer

IF signals up to 2.9 GHz base frequency are sampled in the 1st Nyquist band, while signals from 3.4 GHz up to 3.95 GHz, or from 3.95 GHz up to 5.0 GHz base frequency are sampled in the 2nd Nyquist band. Base frequencies between 2.9 GHz and 3.4 GHz as well as base frequencies higher than 5 GHz are not supported.

7.2 Digitizer instrument properties

These properties apply to all actions within one specification and cannot be changed during a measurement execution. All properties are optional.

7.2.1 connect

This property specifies if the digitizer instrument should connect before the measurement execution. If set to *false*, the instrument is not connected during the measurement.

Default: *true*

Example:

```
connect = true;
```

7.2.2 disconnect

This property specifies whether to disconnect the digitizer instrument after the measurement execution. If set to *false*, the instrument remains connected after the measurement.

Default: *false*

Example:

```
disconnect = true;
```

7.2.3 delay

To fine-tune the synchronization between the digitizer and the other instruments the actual start of an action can be delayed with the delay property. Use this property to adjust the phase of the signal or to add a small delay up to 1 ms. For larger delays up to 10 s and to delay single actions you can use the *wait* statement in the operating sequence.

Default: 0 ns

Example:

```
delay = 9.1ns;
```

7.2.4 superPeriod

This property sets a super period to align all signals in this instrument's phase domain.

Default: Least common multiple (LCM) of all periods.

Example:

```
superPeriod = 103.1ns;
```

7.2.5 [alarm](#)

The WSWB digitizer supports the over-range alarm, which indicates that the captured waveform exceeds the digitizer range. This alarm is configurable and can be turned on or off.

Example:

```
alarm.overrange = enabled;
```

7.2.6 [compensation](#)

Use this property to set the quality of the amplitude compensation. Currently only standard compensation is supported.

Example:

```
compensation = standard;
```

7.3 Digitizer instrument options

The instrument options apply to all actions of a pin or pin group within one specification and cannot be changed during an operating sequence. In the specification file they are located inside the *setup* block together with the instrument properties.

7.3.1 Differential connection

All WSWB channels are single-ended. If the AWG instrument option *differential* is selected, an error message is issued.

7.3.2 IQ option

The WSWB digitizer is an IF measure instrument; the captured IF waveform can either be returned directly as real-valued IF waveform or be down-converted to complex baseband with the setup property *downConversion.iqConversion()*.

The IQ instrument option is ignored.

7.3.3 Internal Cross-Connect

Use the property *internalCrossConnect* to internally connect the AWG output to the digitizer input without a DUT board connection (internal loop back). To establish the cross-connect connection, the option must be set for both the AWG and the digitizer.

Example:

```
config.options = internalCrossConnect;
```

7.3.4 Ground sense

The WSWB instrument does not support ground sensing.

7.3.5 Multi-Site testing

The four digitizer inputs RXA1, RXA2, RXB1 and RXB2 connect to the four RFIM-18 outputs RX AUX A1, RX AUX A2, RX AUX B1 and RX AUX B2. Each of them is driven by a buffer and de-multiplexer with two inputs, so that the WSWB digitizer can serve eight sites in total, one at a time.

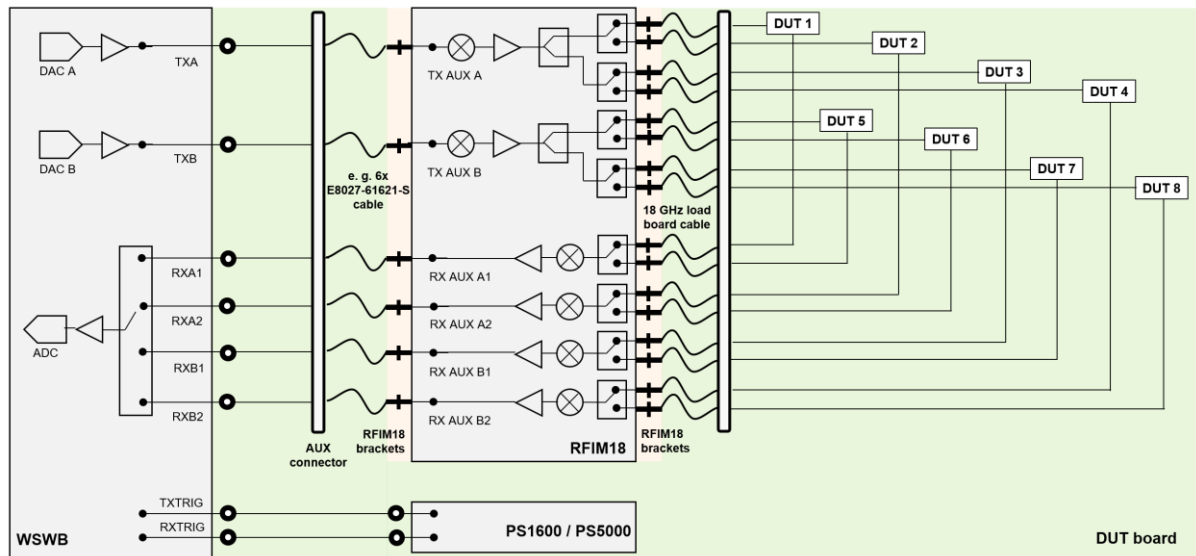


Figure 34: WSWB connected to RFIM-18

Octal site testing is supported without further switches or fan-out buffers on the DUT board. The instrument option *siteInterlacing* is not supported.

7.3.6 Trigger Setup

The WSWB digitizer supports two trigger conditions: *standard* (default) and *transition*.

7.3.6.1 Standard trigger

The standard trigger starts the capturing process at the point in time which is determined by the operating sequence.

7.3.6.2 Transition trigger

The *transition trigger* is a new trigger condition for the digitizer, which triggers on the power of the incoming signal: The digitizer only starts capturing data when the incoming signal exceeds a user defined power threshold. An optional *off-time* can be set to ensure that a signal burst is captured from the very beginning.

The amount of upload data can be significantly reduced with the transition trigger, as it detects the signal bursts. It is only available on the WSWB digitizer.

When the trigger condition *transition* is selected, the digitizer starts sensing the signal power after receiving the arming signal. The capturing process only begins and the waveform data are only stored after the following two conditions have been met:

- the input signal stays below the power threshold for a certain amount of time ("off time", optional)
- the signal power exceeds the power threshold

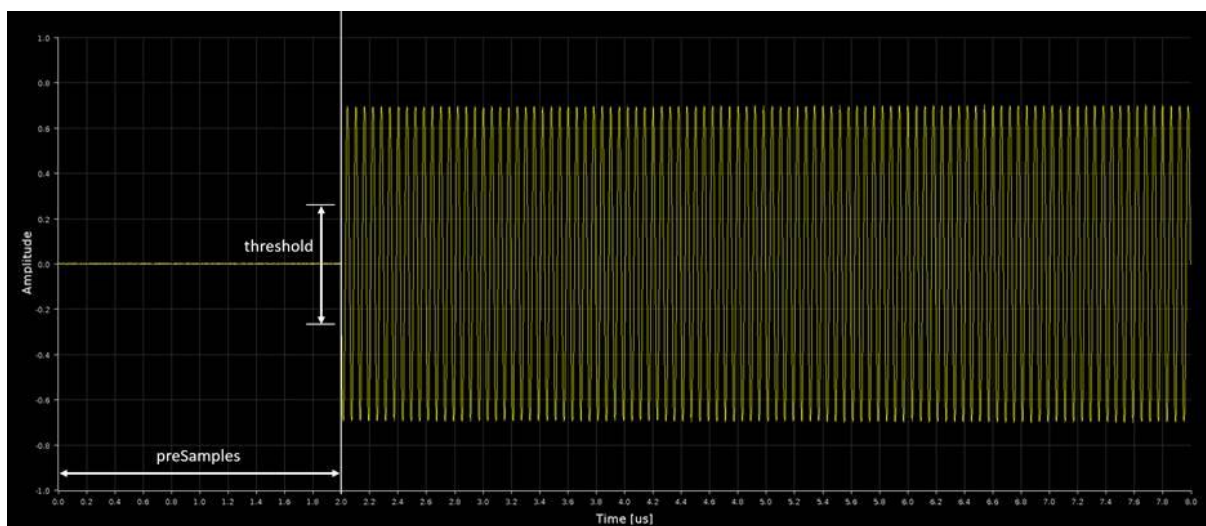


Figure 35: Transition trigger

The trigger is set up in the property group *trigger* of the digitizer setup. It cannot be changed within an operating sequence.

To select the transition trigger mode set `trigger.condition = transition;`

Power threshold

Set up the power threshold for the transition trigger with the property *trigger.threshold()*.

Range: -100 to 14 dBm

Default: -30 dBm

Example:

```
trigger.threshold = -40 dBm;
```

off-time

Set up the off-time (idle time before a signal burst) with the property *trigger.offTime()*.

Range: 0 to 1 s

Default: 0 s

Example:

```
trigger.offTime = 2 us;
```

pre-trigger samples

Use the property *trigger.preSamples()* to set up the number of pre-trigger samples to be uploaded.

Default: 0

Example:

```
trigger.preSamples = 100;
```

timeout

Set up the maximum time to wait for the trigger (trigger timeout) with the property *trigger.timeout()*.

Range: 0 to 1s

Default: 10 ms

Example:

```
trigger.timeout = 20 ms;
```

7.4 Properties of the *measureWaveform* action

Action properties are located inside the *action* block of the specification file. They are only effective for one specific action.

7.4.1 Input Impedance and center ground

The WSWB digitizer has a 50 Ohm terminated single-ended input. An error is returned if the property *inputImpedance* is set to a value other than 50 Ohms. The property *centerGnd* is ignored.

Example:

```
inputImpedance = 50 Ohm;
```

7.4.2 Voltage Range

The input voltage range of the digitizer is determined by the SmarTest software according to the optional digitizer property *expectedAmplitude* or *expectedMaxPower*; the unit is either Volts peak (V_p) or dBm. A reduction of the expected maximum power increases the resolution of the A/D converter and reduces the noise floor.

Default: 14 dBm

Example:

```
expectedAmplitude = 625 mV;
```

The dynamic performance of the digitizer is range-dependent, and also the harmonic distortion, but to a smaller degree.

The properties *expectedAmplitude* and *expectedMaxPower* must not be set when *referencedRfAction* is used; in this case it is taken from the *modPower* action setup of the *rfMeas* instrument.

7.4.3 DC Offset

The WSWB digitizer does not support offset voltages; an error is returned if the property *offset* is set a value other than 0.0 V.

7.4.4 Signal bandwidth and anti-aliasing filter

Use the property *lowPassFilterBandwidth* to set the bandwidth of your signal. This property is mandatory. The SmarTest software checks if the entire spectrum of the expected signal is located inside the pass band of the anti-aliasing filter.

Example:

```
lowPassFilterBandwidth = 800 MHz;
```

The property *lowPassFilterBandwidth* must not be set when *referencedRfAction* is used; in this case it is taken from the *modPower* action setup of the *rfMeas* instrument.

The WSWB digitizer has three different anti-aliasing filters; SmarTest selects the filter based on the base frequency setting.

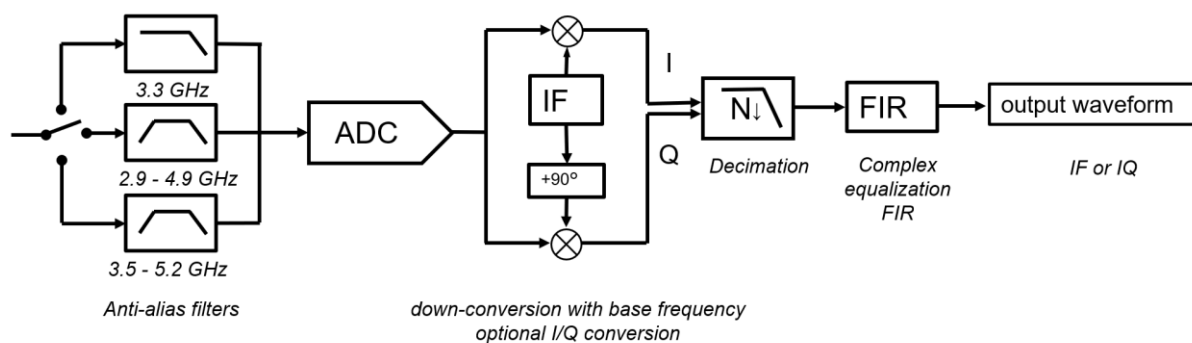


Figure 36: The sampling concept of the digitizer

IF signals with base frequencies up to 2.9 GHz base frequency are filtered by a 3.3 GHz low pass filter and sampled in the 1st Nyquist band.

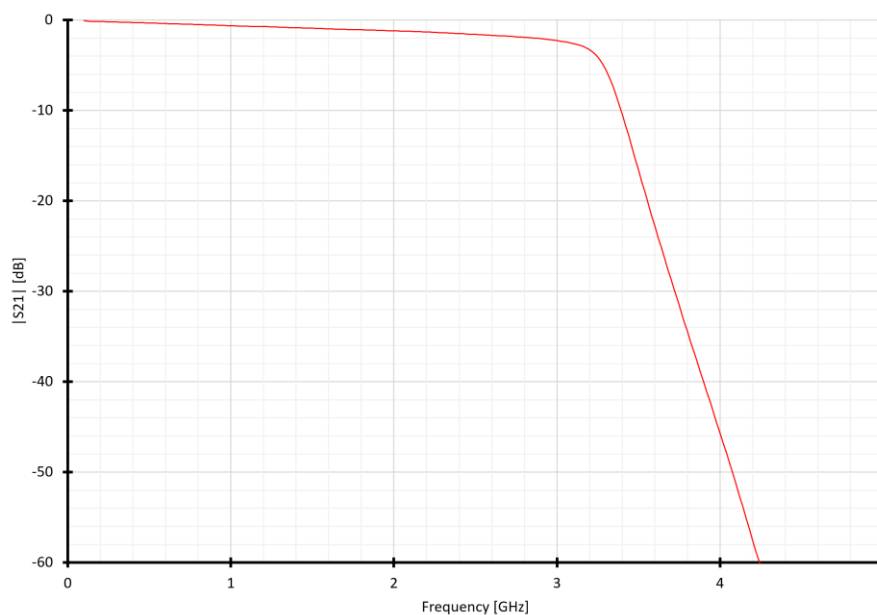


Figure 37: Insertion loss of the 3.3 GHz anti-aliasing low pass filter

Base frequencies between 2.9 GHz and 3.4 GHz and base frequencies higher than 5 GHz are not supported. IF signals with base frequencies from 3.4 GHz up to 3.95 GHz are filtered by a band pass filter with a pass band that ranges from 2.9 to 4.9 GHz and sampled in the 2nd Nyquist band.

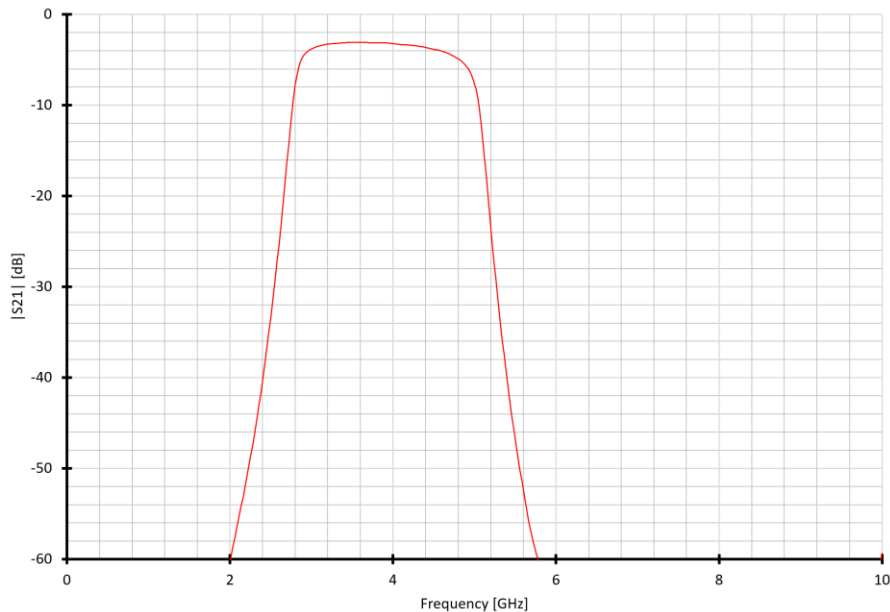


Figure 38: Insertion loss of the 2.9 GHz to 4.9 GHz anti-aliasing band pass filter

IF signals from 3.95 GHz up to 5 GHz are filtered by a band pass filter with a pass band that ranges from 3.5 to 5.2 GHz and also sampled in the 2nd Nyquist band.

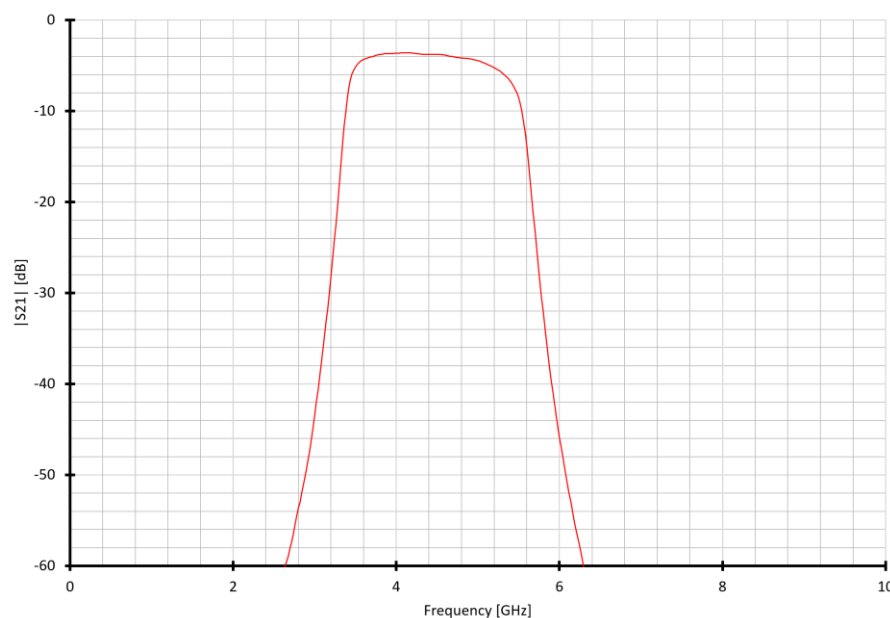


Figure 39: Insertion loss of the 3.5 GHz to 5.2 GHz anti-aliasing band pass filter

7.4.5 Base frequency

The WSWB digitizer input signal is a carrier signal at the *base frequency*, modulated by the complex baseband waveform. Use the mandatory property *baseFrequency* to set this carrier signal frequency. The WSWB digitizer supports base frequencies from 545 MHz to 2.9 GHz and from 3.4 GHz to 5.2 GHz. SmarTest selects the anti-aliasing filter based on the base frequency setting. If *downconversion* is enabled, the captured waveform is down-converted by the base frequency.

The property *base frequency* must not be set when *referencedRfAction* is used; in this case it is taken from the *modPower* action setup of the *rfMeas* instrument.

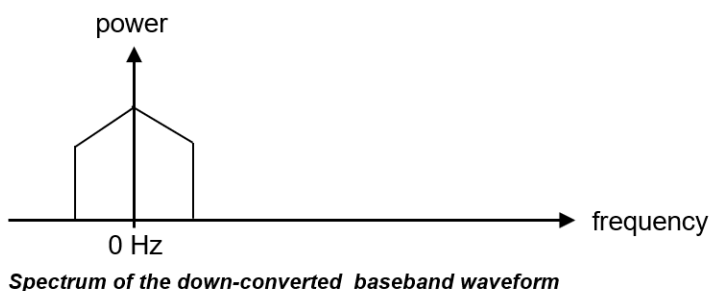
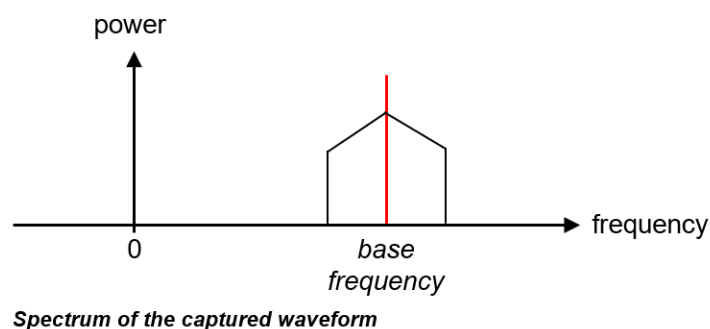


Figure 40: IF spectrum and baseband spectrum

7.4.6 Samples

Use the property *samples* to set the number of samples to be returned by the result accessor. This property is mandatory. The A/D converter of the digitizer might capture a lot more samples in case decimation or averaging is turned on. The maximum number of samples is 10000000.

Example:

```
samples = 98304;
```

7.4.7 Sample Rate

Use the property *sampleRate* to set the sample rate of the waveform to be returned by the result accessor. This property is mandatory. The A/D converter of the digitizer might capture at a higher rate in case decimation or averaging is turned on.

Example:

```
sampleRate = 983.04 MHz;
```

The WSWB digitizer supports the following sample rates:

2G / 4G LTE / 5G NR

- 30.72, 61.44, 122.88, 245.76, 491.52 & 983.04 Msps
- 2611.2 & 5222.4 Msps (3.4 - 3.95 GHz base frequency only)
- 2949.12 & 5898.24 Msps (3.95 – 5.0 GHz base frequency only)
- 7864.32 Msps (500 MHz - 2.9 GHz base frequency only)

IEEE 802.11ad and IEEE 802.11ay (WiGig)

- 1760 & 2640 Msps
- 5280 Msps (3.4 - 3.95 GHz base frequency only)
- 6000 Msps (3.95 – 5.0 GHz base frequency only)
- 7920 Msps (500 MHz - 2.9 GHz base frequency only)

IEEE 802.15.4 (Ultra-Wideband)

- 499.2 & 998.4 Msps
- 5491.2 Msps (3.4 - 3.95 GHz base frequency only)
- 5990.4 Msps (3.95 – 5.0 GHz base frequency only)
- 7987.2 Msps (500 MHz - 2.9 GHz base frequency only)

7.4.8 InputCapacitance

The WSWB digitizer does not support the property *inputCapacitance*. If set, an error is returned.

7.4.9 Infinite

The WSWB digitizer does not support the property *infinite*. If set, an error is returned.

7.4.10 Decimation

Decimation is the process of reducing the sampling rate of a sampled signal. Decimation utilizes filtering to mitigate aliasing distortion, which can occur when simply down-sampling a signal. Prior to decimation the signal is oversampled. This technique is commonly used to reduce digitizer noise and aliases in the captured signal. Decimation only reduces the noise which is produced by the digitizer, and not the noise of the incoming signal itself; therefore it is legitimate to use decimation for noise measurements.

Set the property *decimation* to turn on decimation (*decimation = true*). It is not necessary to modify the sampling setup; SmarTest itself computes the optimum oversampling and decimation setup.

When decimation is turned on, the usable bandwidth of the obtained spectrum might be reduced as the upper part of the spectrum might not be compensated and may contain aliases. Within the usable bandwidth, however, all amplitudes are precise and aliases are suppressed.

The WSWB digitizer samples at sample rates between 5222.4 and 7987.2 Msps. SmarTest uses decimation according to the sample rate setup of the captured data set up by the user. For WSWB digitizer sample rates up to 2949.12 Msps decimation must be turned on (*decimation = true*). For digitizer sample rates higher or equal than 5222.4 Msps decimation can be turned off (*decimation = false*).

Example:

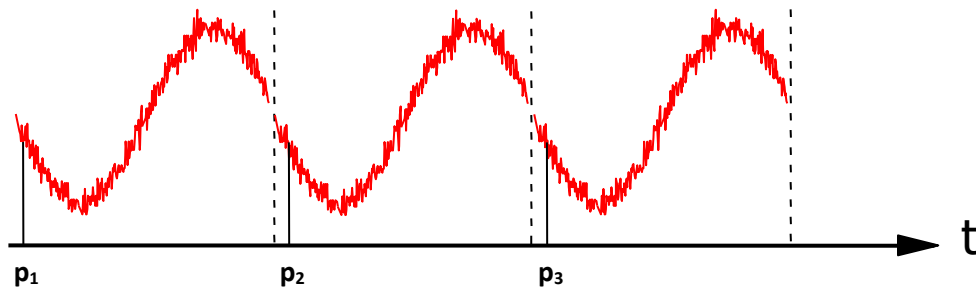
```
decimation = true;
```

7.4.11 BandwidthOfInterest

The WSWB digitizer *measureWaveform* action ignores the setting of the instrument property *bandwidthOfInterest*.

7.4.12 Waveform Averaging

Signals are usually captured in the presence of a significant amount of noise. The WSWB digitizer offers waveform averaging (ensemble averaging) to reduce noise in dynamic measurements. The captured vector is divided into n portions of equal length. Then the mean value of the corresponding data points is determined. Ensemble averaging reduces the noise of the signal source (DUT) and the digitizer; for this reason this method is not appropriate for noise measurements.



$$p = \frac{1}{3} (p_1 + p_2 + p_3)$$

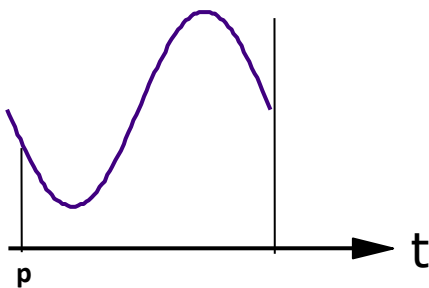


Figure 41: Waveform averaging

Use the action property *waveformAveraging* to average entire waveforms.

Example:

```
waveformAveraging = 3;
```

WSWB does not support averaging per sample (*averagesPerSample*) to average subsequent samples and returns an error if set.

7.4.13 Digital filter

The WSWB digitizer ignores the settings of the instrument property group *digitalFilter*.

7.4.14 DownConversion

Digital down-conversion converts a digitized, real-valued, band limited signal down to a real-valued or complex, lower frequency signal. A numerical oscillator (NCO) generates a complex sinusoid at the base frequency. The input waveform is then multiplied with this sinusoid, resulting in a complex waveform, with images of the original waveform centered at the sum and difference frequency.

The WSWB digitizer channel contains a built-in downconverter, which multiplies the captured samples with the output data of a numerical oscillator running at the base frequency, and splits the IF data into the in-phase (I) and the quadrature (Q) data, or simply shifts the captured data into the baseband location.

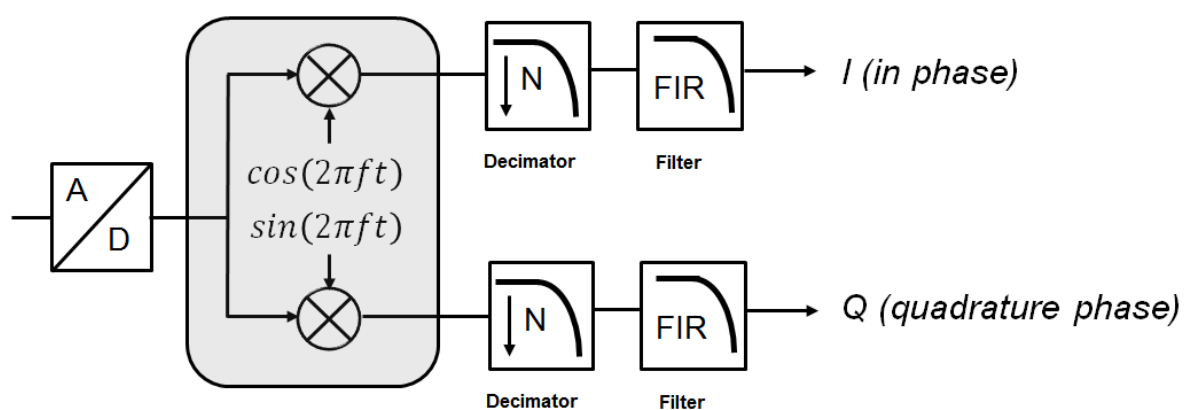


Figure 42: Digital down-conversion with decimator and filter

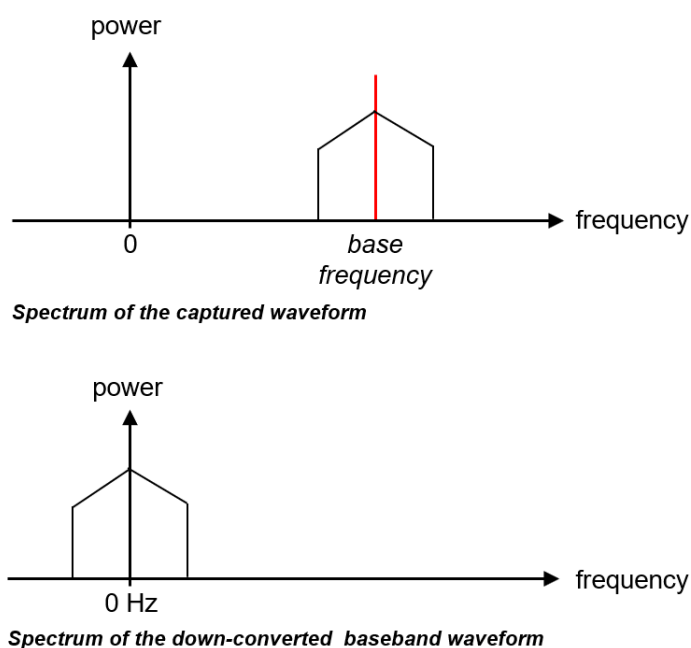


Figure 43: Spectrum of a bandlimited signal modulated on the IF carrier and in baseband

Use the instrument property group *downConversion* to down-convert the captured waveform. The center frequency of the WSWB downconverter is the base frequency and cannot be changed. Set *downConversion.iqConversion = true* to get the captured waveform in complex IQ baseband.

Examples:

```
downConversion() ;  
downConversion.iqConversion           = true;
```

A digital low pass filter after the down-converter passes the difference frequency and rejects the sum frequency image.

Depending on the property *sampleRate*, the resulting complex baseband representation of the original signal is decimated to a lower sample rate without losing any information contained in the original signal.

7.4.15 Target Frequency

This property is not relevant for WSWB and has no effect.

7.4.16 Peak-To-Average Power Ratio

Use the property *peakToAvgPowerRatio* to set the peak-to-average power ratio (PAPR) of the captured waveform in decibels (dB).

Default: 0 dB

Example:

```
peakToAvgPowerRatio = 11.95 dB
```

The property *peakToAvgPowerRatio* must not be set when *referencedRfAction* is used; in this case it is taken from the *modPower* action setup of the *rfMeas* instrument. Please note that also in this case you need to enter the PAPR value of the IF waveform according to the “carrier approach”.

7.4.16.1 PAPR of real-valued waveforms

The peak-to-average power ratio or PAPR is the peak power (peak amplitude squared) of a waveform divided by the average power, and is usually expressed in decibels. It indicates how extreme the peaks are in a waveform. The PAPR is the square of the crest factor, and when expressed in decibels, the PAPR and the crest factor are equivalent. The value can range from 0 dB for DC waveforms to up to 20 dB for multi-tones with a disadvantageous choice of phases.

7.4.16.2 PAPR of complex waveforms

The PAPR of a complex IQ baseband waveform is that of the magnitude of the complex waveform. As a consequence, the PAPR of a complex sine wave is 1.0 or 0 dB, in contrast to the PAPR of a real-valued sinusoidal signal, which is $\sqrt{2}$ or 3.01 dB. The signal analyzer tool provides the convenient function *Analysis* → *Signal Information* to compute the PAPR of real- and complex-valued waveforms.

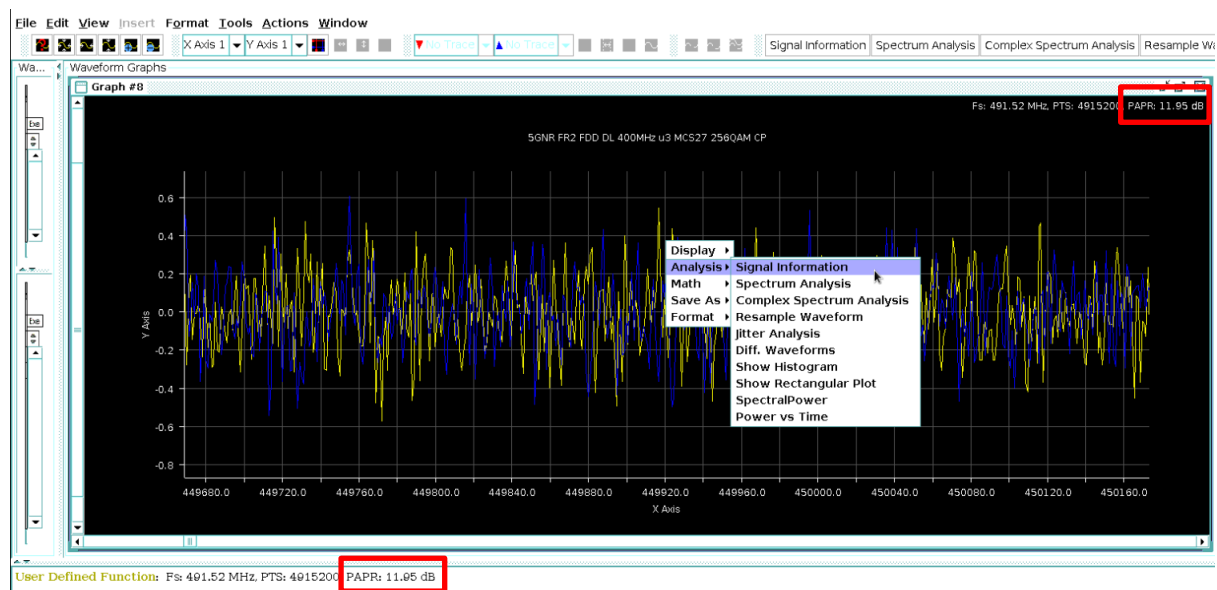


Figure 44: Computing the PAPR of a waveform with the signal analyzer tool

7.4.16.3 *PAPR of modulated waveforms*

For a modulated RF signal, we can determine two different PAPR values that differ by 3.01 dB. The first way is to compute the PAPR of the modulation waveform as described above. This value is equivalent to the ratio of the peak value of the modulation envelope to its RMS value and is sometimes called the "envelope approach". The disadvantage of this approach is that it does not take the carrier into account.

The second approach to compute the PAPR of a modulated RF signal divides the highest amplitude peak that occurs in the modulated carrier signal by its RMS value. This approach takes the RF carrier into account and is usually called the "carrier approach". The PAPR determined in this manner is higher by the magnitude of the crest factor for the sinusoidal carrier compared to the carrier approach. This approach is important when setting the input range of an RF or IF measurement instrument.

Or, in other words: When a complex waveform is modulated onto an IF or RF carrier, the PAPR of the modulated signal is the PAPR of the complex waveform plus the PAPR of the sinusoidal carrier, which is 3.01 dB.

7.4.16.4 *PAPR and input range setting*

The property *peakToAvgPowerRatio* is important for the input range setting of the digitizer which needs to be calculated according to the expected peak power. The *expectedMaxPower* is an average power value, and therefore the PAPR of the expected waveform needs to be taken into account to avoid clipping.

Note: The WSWB digitizer always captures IF waveforms, independent of whether the returned waveform is in IF or complex IQ format. The property *peakToAvgPowerRatio* is the PAPR value of the IF waveform according to the "carrier approach". If the returned waveform is a down-converted, complex IQ waveform, you need to add 3.01 dB to the PAPR of the down-converted IQ waveform.

7.4.17 ReferencedRfAction

The WSWB digitizer is the wideband IF capture instrument of the RFIM-18. The power, base frequency and bandwidth settings need to be synchronized between these two instruments. The easiest way to accomplish this is by making use of the property *referencedRfAction*. The argument of the property *referencedRfAction* is the name of the corresponding *rfMeas* action.

The *measureWaveform* properties *expectedAmplitude*, *expectedMaxPower*, *base frequency*, *peakToAvgPowerRatio* and *bandwidthOfInterest* must not be set when *referencedRfAction* is used.

Example:

```
referencedRfAction = "rxalAction";
```

The choice of digitizer sample rates is reduced when setting up the *referencedRfAction*. Sample rates up to 1760 Msps and 2640 Msps are available for all base frequencies; the availability of other sample rates depends on the base frequency, which is automatically selected by SmarTest in this case.

7.5 Digitizer Setup with the Device Setup API

In addition to the SmarTest Setup Format (SSF), SmarTest's own domain specific language, AWGs and digitizers can also be set up using the Device Setup API, the Java test method interface to the instruments. The Device Setup API provides a complete set of functions to set up measurements. The example below shows a complete digitizer setup.

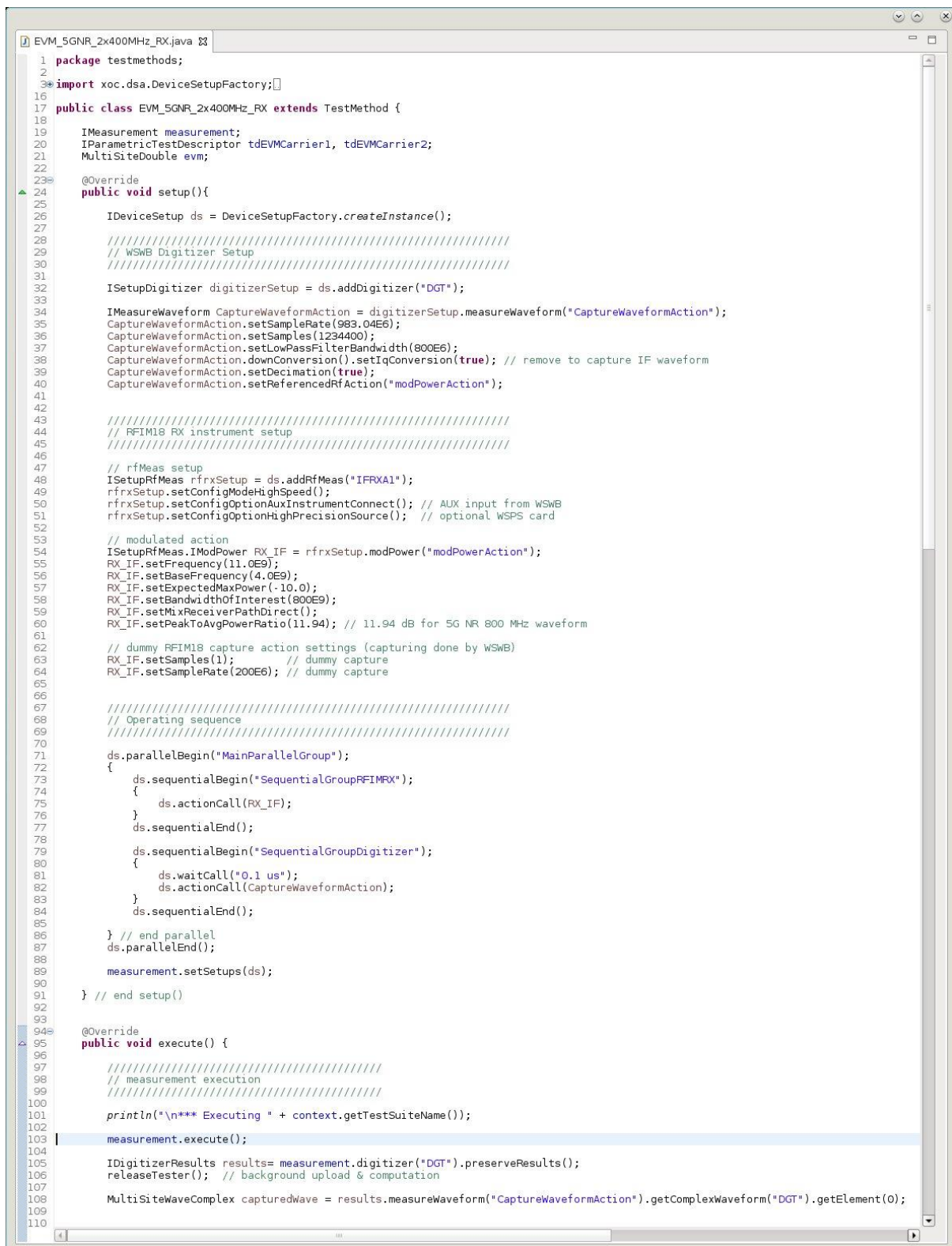


Figure 45: WSWB digitizer setup for a 5G NR EVM test

```

93
94 @Override
95 public void execute() {
96
97     ///////////////////////////////////////////////////
98     // measurement execution
99     ///////////////////////////////////////////////////
100
101     println("\n*** Executing " + context.getTestSuiteName());
102
103     measurement.execute();
104
105     IDigitizerResults results= measurement.digitizer("DGT").preserveResults();
106     releaseTester(); // background upload & computation
107
108     MultiSiteWaveComplex capturedWave = results.measureWaveform("CaptureWaveformAction").getComplexWaveform("DGT").getElement(0);
109
110
111     ///////////////////////////////////////////////////
112     // display time-domain waveform and down-converted spectrum
113     ///////////////////////////////////////////////////
114
115     capturedWave.plot("time-domain waveform");
116     capturedWave.spectrum(SpectrumUnit.dBm).plot("power spectrum");
117
118
119     ///////////////////////////////////////////////////
120     // Demodulation
121     ///////////////////////////////////////////////////
122
123     int zeros[] = {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0};
124
125     // demodulation settings
126     Demodulation demod = new Demodulation("5GNRUpLink");
127
128     demod.setInputParameter("5NRApplyFreqErrorCorrection", 1); // boolean
129     demod.setInputParameter("5NRApplyMultiCarrierFilter", 1); // boolean
130     demod.setInputParameter("5NRBurstMode", 1);
131     demod.setInputParameter("5NRBurstSearchAboveNoiseFloorThreshold", 10.000000);
132     demod.setInputParameter("5NRBurstSearchThreshold", -20.000000);
133     demod.setInputParameter("5NRCarrierOffsetHz", -200000000.000000);
134     demod.setInputParameter("5NRDCPunctured", 1);
135     demod.setInputParameter("5NRDMRSAddPos", 0);
136     demod.setInputParameter("5NRDMRSConfiguration", 1);
137     demod.setInputParameter("5NRDMRSDuration", 1);
138     demod.setInputParameter("5NRDMRSPowerBoostdB", 0.000000);
139     demod.setInputParameter("5NRDMRSTypeAPos", 2);
140     demod.setInputParameter("5NRDataBurstTestConfig", 103);
141     demod.setInputParameter("5NREVMIncludeInactive", 0); // boolean
142     demod.setInputParameter("5NREVMIncludePXCCH", 1); // boolean
143     demod.setInputParameter("5NREVMIncludePXSCH", 1); // boolean
144     demod.setInputParameter("5NREVMIncludeRS", 1); // boolean
145     demod.setInputParameter("5NRLeastRhoForDemodulation", 0.700000);
146     demod.setInputParameter("5NRMaxEVMToSaveReference", 2.500000);
147     demod.setInputParameter("5NRMeasIntInSlots", 1);
148     demod.setInputParameter("5NRMultiCarrierFilterBw", 1000000.000000);
149     demod.setInputParameter("5NRMultiCarrierFilterTransientBw", 9850000.000000);
150     demod.setInputParameter("5NRNumology", 4);
151     demod.setInputParameter("5NRPushMapping", 1);
152     demod.setInputParameter("5NRResLenInSlots", 1);
153     demod.setInputParameter("5NRRLRemoveIQOffset", 1);
154     demod.setInputParameter("AutoDetectParameters", 0); // boolean
155     demod.setInputParameter("Bandwidth", 16);
156     demod.setInputParameter("CarrierFrequencyHz", 3750000000.0);
157     demod.setInputParameter("CellID", 0); // int
158     demod.setInputParameter("DCFrequencyOffsetHz", 0.000000); // double
159     demod.setInputParameter("DCOffsetAtCenterOfChannel", 1); // boolean
160     demod.setInputParameter("DMRSMode", 0);
161     demod.setInputParameter("DemodConditions", 0);
162     demod.setInputParameter("EqualizerTraining", 3);
163     demod.setInputParameter("ErrorFrequencySpectrum", 0); // boolean
164     demod.setInputParameter("OFDMType", 0);
165     demod.setInputParameter("OperatingBand", 77);
166     demod.setInputParameter("ResourceBlocksMaxNumber", 0);
167     demod.setInputParameter("ResourceBlocksOffset", 47);
168     demod.setInputParameter("ResourceBlocksUsed", 37);
169     demod.setInputParameter("SampleRate", 983.04E6);
170     demod.setInputParameter("SmoothEqualizer", 0); // boolean
171     demod.setInputParameter("SymbolTimeAdjustment", -3.125000);
172     demod.setInputParameter("SyncSearchLengthSym", 4);
173     demod.setInputParameter("TrackAmplitude", 1); // boolean
174     demod.setInputParameter("TrackPhase", 1); // boolean
175     demod.setInputParameter("TrackTiming", 1); // boolean
176     demod.setInputParameter("halfSlotDemodEnable", 0); // boolean
177     demod.setInputParameter("dmrsSequenceVP", zeros); // vector
178
179

```

Figure 46: Demodulation setup for a 5G NR EVM test (example)

```

179
180 // Console print out and data log
181 // Console print out and data log
182 // Console print out and data log
183
184 int carrier = -1;
185 double CarrierOffset;
186
187 // data log setup
188 tdEVMCarrier1.setTestName("5G NR EVM Carrier 1").setTestText("5G NR EVM Carrier 1").setTestNumber(100);
189 tdEVMCarrier1.setLowLimit(0.5).setHighLimit(2.0);
190 tdEVMCarrier2.setTestName("5G NR EVM Carrier 2").setTestText("5G NR EVM Carrier 2").setTestNumber(200);
191 tdEVMCarrier2.setLowLimit(0.5).setHighLimit(2.0);
192
193 for (int isite: context.getActiveSites())
194 {
195     println( " \n===== SITE " + isite + " =====" );
196
197     // demodulate both 400 MHz carriers
198     for (carrier=1; carrier<=2; carrier++)
199     {
200         CarrierOffset = carrier*400E6-600E6;
201         demod.setInputParameter("5GNRCarrierOffsetHz", CarrierOffset);
202         demod.execute(capturedWave);
203
204         println(String.format( "Carrier Offset:          %2.0f MHz", CarrierOffset/1E6));
205
206         println( " \n*** Frequency and quadrature error ***" );
207         println(String.format( "Frequency error:      %2.1f          Hz", demod.getResultParameterDouble("FrequencyError").get(isite)));
208         println(String.format( "IQ gain imbalance:    %2.8f dB", demod.getResultParameterDouble("IQGainImbalance").get(isite)));
209         println(String.format( "IQ offset:            %2.1f          dB", demod.getResultParameterDouble("IQOffset").get(isite)));
210         println(String.format( "IQ quad error:        %2.5f          deg", demod.getResultParameterDouble("IQQuadError").get(isite)));
211         println(String.format( "IQ timing skew:       %2.3e          s", demod.getResultParameterDouble("IQTimingSkew").get(isite)));
212
213         println( " \n*** EVM rms ***" );
214         println(String.format( "EVM rms (total):      %2.2f %%", demod.getResultParameterDouble("EvmRms").get(isite)));
215         println(String.format( "EVM rms data:         %2.2f %%", demod.getResultParameterDouble("EvmRmsData").get(isite)));
216         println(String.format( "EVM rms pilot:        %2.2f %%", demod.getResultParameterDouble("EvmRmsPilot").get(isite)));
217
218         if (carrier==1)
219         { tdEVMCarrier1.evaluate(demod.getResultParameterDouble("EvmRms")); }
220
221         else
222         { tdEVMCarrier2.evaluate(demod.getResultParameterDouble("EvmRms")); }
223
224     } // end carrier loop
225
226 } // end site loop
227
228 println("\n*** " + context.getTestSuiteName() + " finished\n");
229
230 }
231 }
232

```

Figure 47: Evaluation and data log setup for a 5G NR EVM test (example)

7.6 Modulated power

Modulated power can be measured by computing the RMS power of a captured waveform, subtracting the RFIM-18 gain and adding the DUT board losses. An example is shown below; note that the action and pin name must be set correctly.

```
// compute modulated power
MultiSiteDouble loss    = new MultiSiteDouble(PowerLoss);
MultiSiteDouble ten     = new MultiSiteDouble(10.0);
MultiSiteDouble gain    =
measurement.rfMeas("IFRXA1").modPower("modPowerAction").getExpectedMaxPowerAtAuxPort().subtract(measurement.rfMeas("IFRXA1").modPower("modPowerAction").getExpectedMaxPower());

// modulated signal power
MultiSiteDouble power =
capturedWave.abs().rms().pow(2).multiply(ten).log10().multiply(10)
.subtract(gain).add(loss);
```

Figure 48: Computation of modulated power

The multi-site double *power* contains the modulated powers of all sites. Additional power losses caused by wiring, traces or sockets are included by *loss* (site-specific) or *PowerLoss* (site common).

8. Synchronization

8.1 Preparation times

All analog actions need to be prepared before they can be executed: signal paths and clock frequencies need to be switched, the registers of the converters need to be reprogrammed, digital filter parameters need to be loaded, etc. During these preparation times, the execution of actions is not possible. The amount of preparation time is displayed in the operating sequence view by the horizontal black bars before the action.

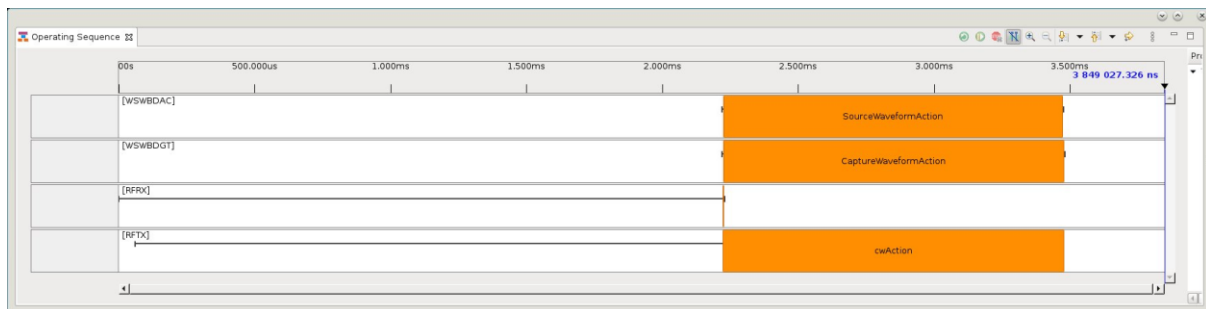


Figure 49: Actions in an operating sequence with preparation times

These preparation times do neither include filter settling times, nor transient responses of the DUT or the DUT board. Take them into account by extending waveforms and removing samples which have been captured during transition times.

8.2 Delay

Actions can be delayed with the *delay* property, with the *wait* statement inside the operating sequence or by attaching the action to a different vector in the pattern.

8.3 Sync Accuracy

SmarTest synchronizes action starts as precise as possible for the given analog sample rates, digital waveform periods, and wait times. This can lead to long synchronization times or even to conflicts, which SmarTest cannot resolve.

Coherently captured, periodic analog waveforms usually do not require precise synchronization with other instruments, so that in many cases these conflicts can be resolved, or sync times can be shortened by relaxing the sync accuracy to a value in the range of tens or hundreds of nanoseconds.

syncAccuracy only relaxes the absolute synchronization accuracy; even with relaxed requirements the actions are started in a deterministic manner.

By default action starts are synchronized with a default accuracy of 1 ns.

8.4 Synchronization of WSWB and the RF subsystem

The WSWB digitizer is the wideband IF capture instrument of the RFIM-18. To capture a wideband modulated RF signal, a WSWB *measureWaveform* action and an RF *modPower* action have to be set up. The power, base frequency and bandwidth settings can be synchronized between these two instruments with the property *referencedRfAction*.

It is recommended to run the WSWB and the corresponding WSRF action in a parallel group.

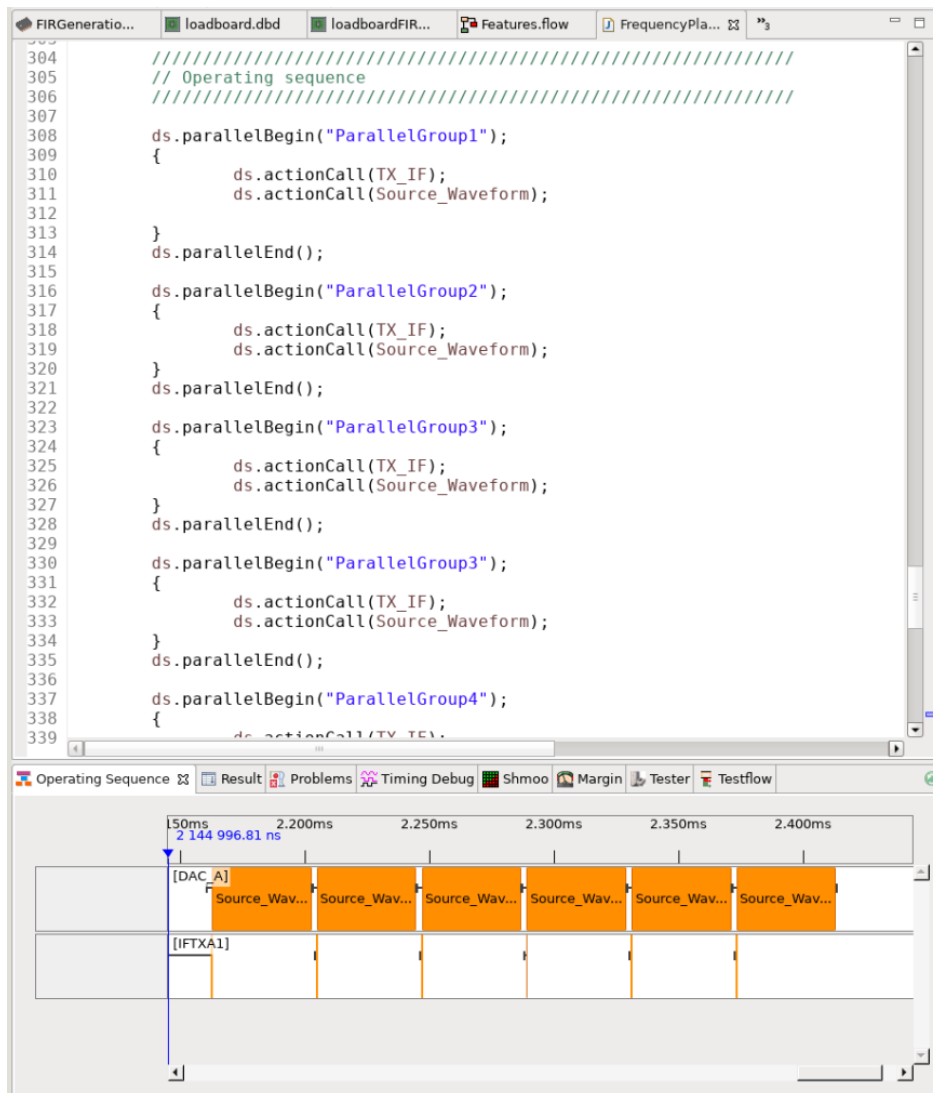


Figure 50: WSWB and WSRF action in a parallel group

9. Calibration and Diagnostics

9.1 Prerequisites

Calibration and diagnostics of WSWB require a WSWB calibration board E9750A-001. This DUT board provides the signal connections to the corresponding RFIM18 module, which is used as a reference instrument. It is not possible to run WSWB maintenance without the RF subsystem including an RFIM-18.

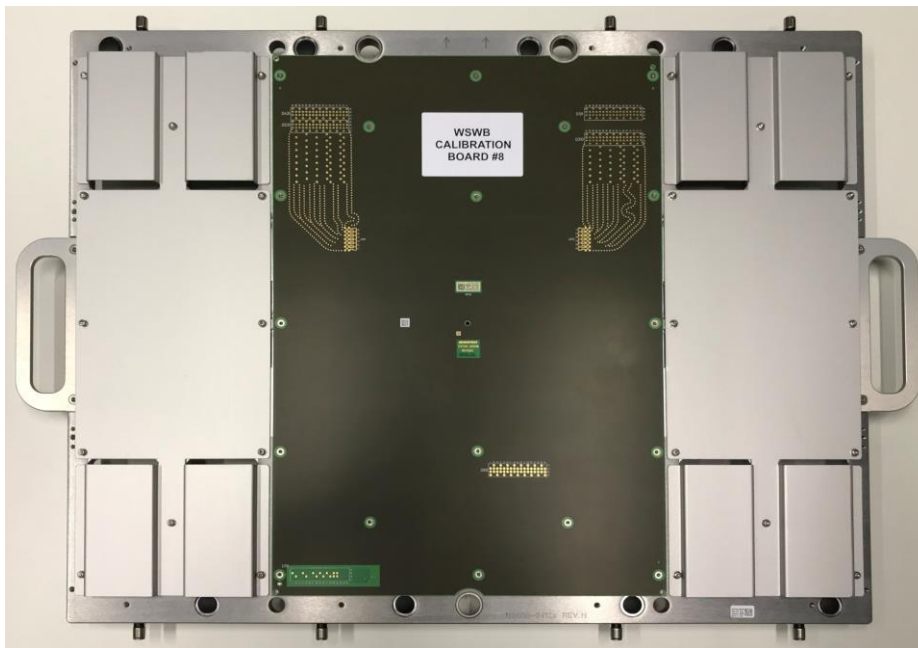


Figure 51: WSWB Calibration Board E9750A-001

9.2 WSWB maintenance as part of RF maintenance

Calibration and diagnostics are started from the control desk of the SmarTest8 maintenance perspective, a dedicated Eclipse perspective which includes an overall maintenance status view. WSWB is included in the section *RF Auxiliary AWG and Digitizer* of the *RF maintenance activities*. The user will be prompted to dock the WSWB calibration board at the appropriate time.

It is recommended to perform diagnostics before calibration to identify potential hardware issues beforehand.

Calibration data are stored in the calibration database of the tester workstation. The calibration is valid for the standard maintenance period of three months. Data from previous calibration runs can be restored via the pull-down menu *Maintenance -> Compensation Data*.

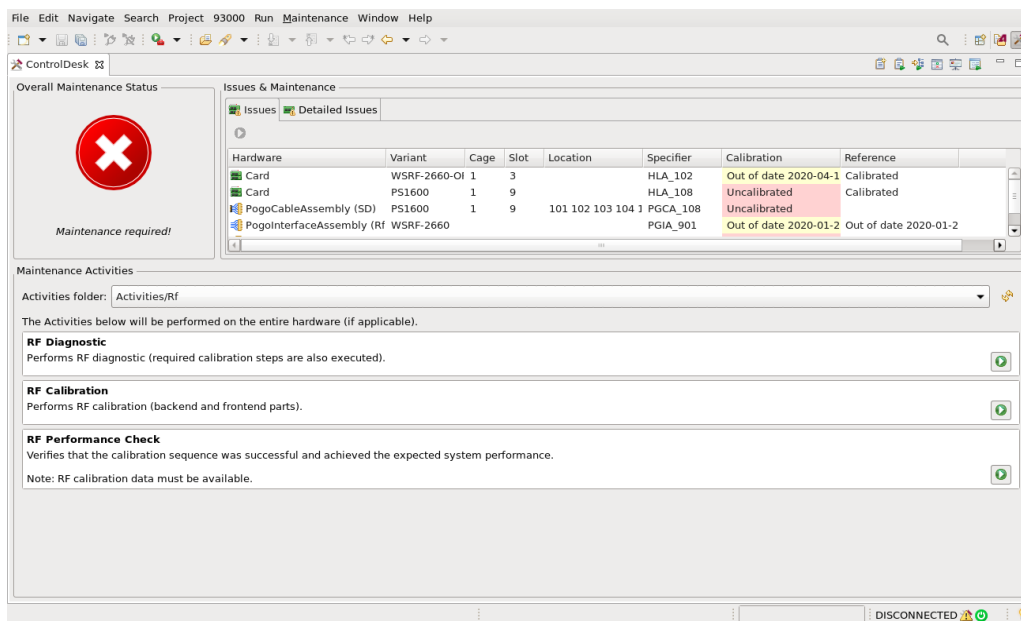


Figure 52: RF maintenance activities accessible from the control desk

9.3 Maintenance of a single WSWB card

If a WSWB card is added to 93000 test system or exchanged, there is no need to perform the complete RF diagnostic and calibration. To calibrate WSWB only, open the menu *Maintenance -> Activity-based Setup*, uncheck all hardware except WSWB in the hardware tree, and then run the desired RF activity.

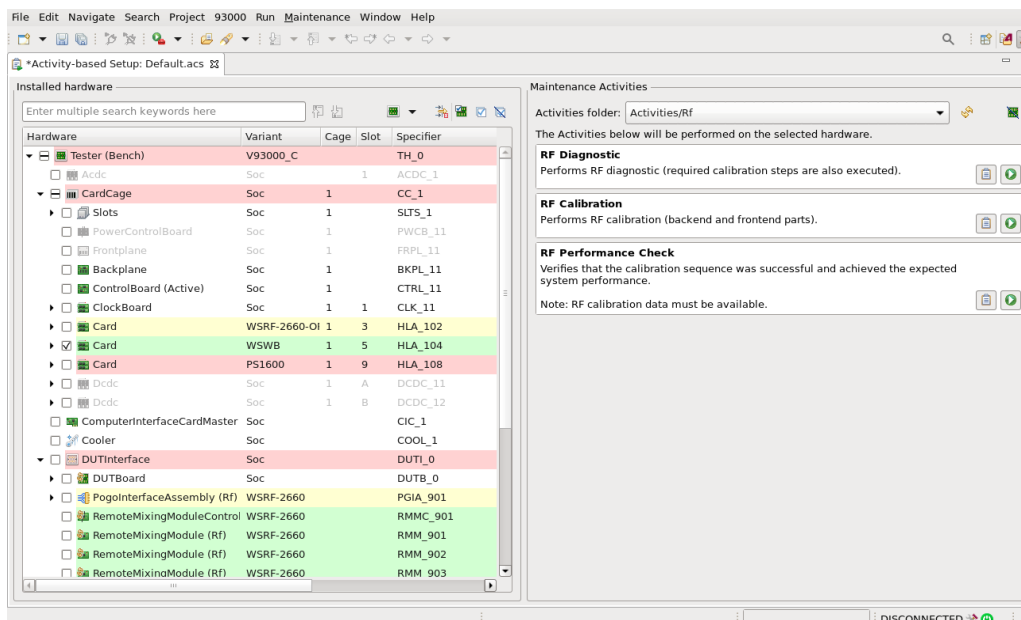


Figure 53: Selecting WSWB in an activity-based setup

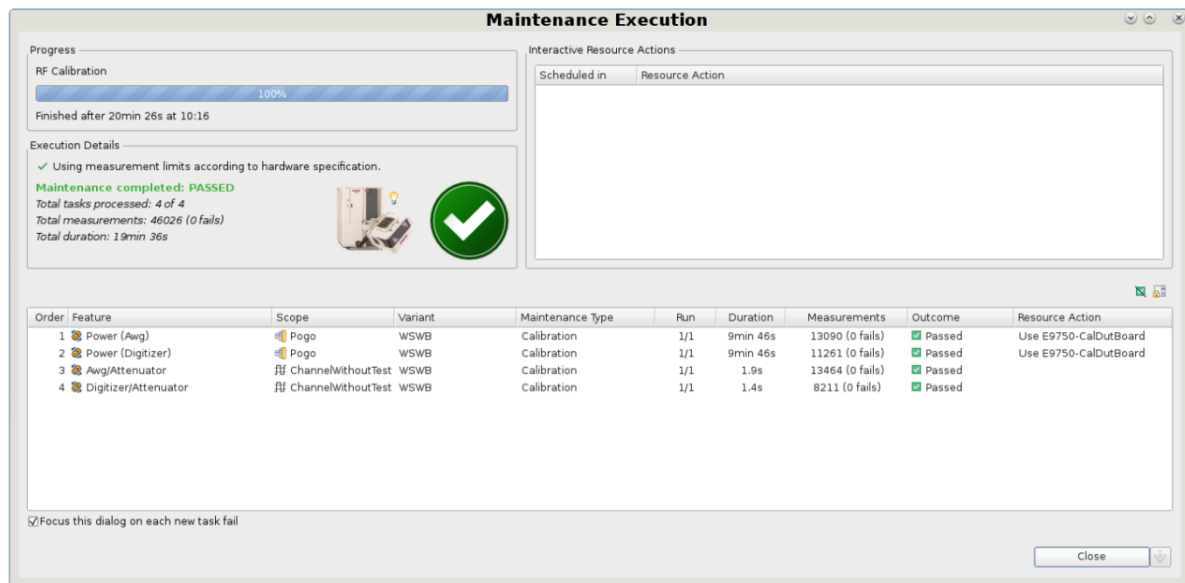


Figure 54: WSWB Maintenance Execution window

9.4 RF signal path and calibration plane

The WSWB calibration board E9750A-001 connects the WSWB IF signals to the corresponding auxiliary (AUX) ports of the RFIM-18 module, and the RFIM-18 RF in- and outputs are connected in a loopback configuration.

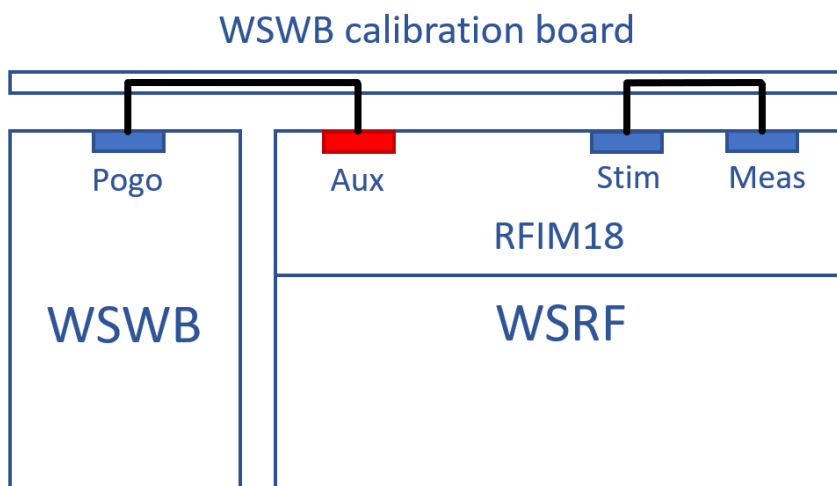


Figure 55: Calibration plane at the Aux port

RFIM-18 RF stim pogo number	RFIM-18 RF stim port name	RFIM-18 RF meas pogo number	RFIM-18 RF meas port name
90313	TXA1(1)	90403	RXB2(1)
90314	TXA1(2)	90402	RXB1(2)

90315	TXA2(1)	90401	RXB1(1)
90316	TXA2(2)	90404	RXB2(2)
90413	TXB1(1)	90303	RXA2(1)
90414	TXB1(2)	90302	RXA1(2)
90415	TXB2(1)	90301	RXA1(1)
90416	TXB2(2)	90304	RXA2(2)

Table 7: RFIM-18 RF loopback pogo routing of the WSWB Calibration Board E9750A-001

The calibration plane for WSWB are the AUX ports of the RFIM-18 module, the path loss of the connection between the WSWB IF ports and the RFIM-18 AUX inputs through the calibration board is included in the calibration data. The best power accuracy is achieved if the WSWB to RFIM-18 connection on the DUT board has a similar path loss as the connection on the calibration board.

10.Licenses

The following licenses are available for the Wave Scale Wideband card *E9750A*:

Part Number	Description
N9750-STIM-PRM	License for the Wave Scale Wideband AWG, perpetual
N9750-MEAS-PRM	License for the Wave Scale Wideband digitizer, perpetual

Table 8: Licenses for E9750A (WSWB)

11. Performance Optimization

11.1 Losses

All power losses in the drive path have to be compensated by driving higher power levels, which leads to increased signal distortion of the AWG. On the measure side all losses result in lower power levels and thus a reduced signal-to-noise ratio. Therefore it is highly recommended to keep all losses (trace, cable, etc.) as small as possible.

11.2 Calibration

A thorough calibration of the entire RF subsystem including WSWB is a prerequisite for high performance measurements.

11.3 The Wave Scale Precision Source (WSPS) card

The Wave Scale Precision Source (WSPS) card significantly improves the phase noise performance of the WSRF card and is indispensable for EVM and phase noise measurements. It is enabled in a measurement setup with the option *setConfigOptionHighPrecisionSource()*.

11.4 Referenced RF action

To avoid issues and to obtain the best signal performance it is recommended to use *referencedRfAction*. When using *referencedRfAction* SmarTest selects the optimum base frequency and power level for the WSWB/RF18 interface.

11.5 Base frequency

The signal performance depends on the base frequency setting. In most cases the best performance is obtained when using *referencedRfAction*. In other cases varying the base frequency may lead to better signal performance.

11.6 Digitizer input range

The system software selects the appropriate digitizer input range, based on the setup property *expectedPower* or *expectedAmplitude*. When using *referencedRfAction*, SmarTest optimizes the power level on the RF18 / WSWB interface.

The dynamic performance of all digitizers is range-dependent, and also the harmonic distortion, but to a smaller degree. In general it is recommended to set the *expectedPower* or *expectedAmplitude* to a value close to the maximum expected power without much of a margin.

For distortion measurements it can be advantageous to use a higher range than required; this reduces the modulation of the digitizer and reduces distortion at the cost of higher quantization noise.

For RMS EVM measurements it can be advantageous to use a lower input range than required, especially if the waveform contains very few but extreme power peaks (high PAPR). Even though the peaks might clip and the corresponding symbols contain higher EVM, the quantization noise of vast majority of symbols is reduced and the RMS EVM can be lower.

11.7 Stim path optimization

There are a couple of RF stim setup properties, which can be used to significantly improve the signal performance of the stim path. The operating point of the WSWB should be chosen such that provides best performance. The rule of thumb is: the higher the power level the more distortion, and the lower the power level the more noise. To reduce noise increase the power, to reduce distortion reduce the power.

As the output power level of WSWB is defined by the RFIM-18 when using *referencedRfAction*, it cannot be set up directly. RFIM-18, however, can be forced to choose a certain input power level for the WSWB / RFIM-18 interface with the setup properties *aux().setVsgMinPower()* and *aux().setVsgMaxPower()*.

The next step is to optimize the amplifier and attenuator stages of the RFIM-18 for the given objective. Three options are available: *setOptimizationNoiseFloor()* works with high power levels at the instrument's input and shifts attenuation close to the output. *setOptimizationImd()* avoids high power levels to reduce the overall distortion. Use *setOptimizationBalanced()* together with the property *setOptimizationNoiseBandwidth()* for a trade-off between distortion and noise.

Example:

The optimization setting is key for a good EVM performance. For the EVM measurement of a 5GNR signal, the optimum WSWB output power level is usually in the range from -5 to -2 dBm.

```
TX_IF.aux().setVsgMinPower(-5.0);  
TX_IF.aux().setVsgMaxPower(-2.0);
```

Figure 56: Restrict AUX power level to -5 to -2 dBm

5GNR EVM is much more sensitive to noise than to distortion. Minimize the noise level of the stim path with the setup property *setOptimizationNoiseFloor()*.

```
TX_IF.setOptimizationNoiseFloor();
```

Figure 57: Optimize RF instrument setup for minimum noise